Solar Energy 197 (2020) 163–198

Contents lists available at ScienceDirect

Solar Energy

journal homepage: www.elsevier.com/locate/solener

Recent advancements in PV cooling and efficiency enhancement integrating phase change materials based systems – A comprehensive review

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ARTICLE INFO

Keywords:
Solar energy
Photovoltaics
Cooling
Efficiency enhancement
Performance
Phase-change materials
PV/T
Nanofluids
Optical filters

ABSTRACT

In recent years, researchers are fascinated to counter problem of PV-efficiency decline arising from high operating temperatures, especially in hot climates. This article conducts a comprehensive review of research activities performed in last 5 years, on cooling techniques with phase-change materials (PCMs), nanofluids and their combined use, leading to efficiency enhancement. By passive cooling approach with PCMs, it is found that maximum enhancement up to 20% in PV-efficiency can be achieved. Effectiveness of PCM for PV is more prominent in summer than in winter. Incorporations of fins inside PCM container at PV rear, results in much improved heat conduction within PCM. Now-a-days, researchers have grown interest in composite PCMs for PV cooling due to their enhanced thermal conductivity. Moreover, better heat regulation as well as PV-surface temperature uniformity can be achieved with two PCMs at a time having different melting points. Studies suggest that combination of passive & active cooling techniques helps in further lowering of PV-cell temperature, leading to enhancement in PV-efficiency with additional thermal power generation. PV-efficiency of water-based hybrid PV/T systems can be improved by 32% by integration with PCM. Although nanofluid-based PV/T systems have been proved to enhance PV-efficiency by more than 60%, but combined use of PCM & nanofluid is more effective approach for PV cooling than individual use of PCM or nanofluid. If combination is made between nanofluid & nano-PCM, electrical power & efficiency can further be enhanced. Nanofluids can also be considered a good spectral filter alternative as they require small thickness and are able to be tuned by varying nanoparticles conc. Finally, environmental impacts & economic viability of mentioned cooling techniques, were discussed. Studies show that PV/PCM systems become expensive & less feasible when operated in single junction due to long payback period up to 20 years. Economic feasibility can be increased by combining passive & active cooling techniques which can increase system compactness and lower its cost.

1. Introduction

During last two decades, worldwide share of renewable energy as a source of energy and long-term planning for its sustainable development has increased tremendously, as they have been authenticated the most effective & efficient solutions with reduced emissions of greenhouse gases (Dincer, 2000; Ellabban et al., 2014; Panwar et al., 2011; Apergis and Payne, 2010; Wüstenhagen et al., 2007). About $1.8*10^{11}$ MW of solar energy received by earth, is 1000 times more than the power being consumed from all sources making solar energy the most abundant, inexhaustible & eco-friendly source of energy among all renewable energy sources (Sopian et al., 2000; Parida et al., 2011; Sargunanathan et al., 2016). Solar energy coming from sun in the form of heat & light can be used for thermal applications as in thermal collectors, power supply purposes as in photovoltaic etc. (Sharma, 2011; Sundström, 2009; Lewis, 2007). Photovoltaic (PV) systems have solar cells made up of semi-conductors which convert sunlight into electricity through photoelectric effect (Mickey, 1981; Shukla et al., 2017). According to roadmap provided by IEA, capacity of photovoltaic installed throughout the world will be 4600GW and so it will be providing about 16% of total electricity production (International Energy Agency IEA, 2014). Hence, PV will be providing a clean & sustainable tool for power production at a global level (Michael and Goic, 2015).

PV cells have capacity to absorb 80% of solar energy falling on their surface but their electricity conversion is currently only 12% to 18% with record efficiency of 24.7%. Remaining solar energy is converted into heat raising the temperature of PV-cell up to 40 °C above ambient temperature (Kurtz, 2012:; Agrawal and Tiwari, 2010). Various factors affecting the electricity conversion efficiency of PV modules are irradiance, operating temperature, dust, humidity etc. (Katsumata et al., 2011; Ali et al., 2017; Chatterjee and Tamizhmani, 2012; Bashir et al., n.d.). Simon & Meyer (Simon and Meyer, 2010) stated that at high

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https://doi.org/10.1016/j.solener.2019.11.075
Received 21 June 2018; Received in revised form 19 November 2019; Accepted 21 November 2019
Available online 09 January 2020
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Nomenclature

C_p	Specific heat, J/kg K
CNTs	Carbon Nanotubes
A	Area, m^2
P	Electrical Power Output
E	Energy
FF	Fill factor
V	Voltage, Volt
I	Current, Amp
G	Incident radiation, W/m^2
k	Thermal conductivity, W/m K
m	Mass flow rate, kg/s
Re	Reynolds Number
T	Temperature, K
PV	Photovoltaic
PVT or PV/T	Photovoltaic Thermal unit

Greek symbols

η	Energetic Efficiency
ρ	Density (kg/m^3)

Subscripts

f,i	Fluid inlet
f,o	Fluid outlet
el	Electrical
th	Thermal
ref	Reference
c	Collector
pv	Photovoltaic
Pvt	Photovoltaic thermal unit

temperature, module efficiency decreased due to increase in carrier concentrations at the p-n junction. With rising temperature of PV-module, fill factor as well as open circuit voltage reduced leading to drop in electrical efficiency (Ali et al., 2016; Bashir et al., 2014; Bashir et al., n.d.; Amin et al., 2009). King and Eckert (1996) found decrease of 0.2–0.5% electrical efficiency with every 1 ˚C increase in PV-module temperature. Cueto (2002) found efficiency decrease of 1 to 2% across 30 ˚C temperature span of photovoltaic module. Many authors further reported that electrical conversion efficiency of PV reduced by 0.3–0.65%/K (Sandberg, 1999; Krauter et al., 1999).

As mentioned above, high operating temperature of photovoltaic is one of the major problems in electricity production through PV. In this regard, thermally managed PV systems which are more effective & efficient than conventional-PV systems, have got significant attention now-a-days. In these systems, solar energy which is not converted into electricity and wasted as heat, is absorbed in heat storage medium. Different thermal energy storage media include air, water, phase-change materials (PCMs), nanofluids etc. (Xu et al., 2014; Alva et al., 2018). PCMs have been found to provide an attractive solution due to their high heat storage capacity during phase-change process which occurs at constant temperature (Lin et al., 2018; Souayfane et al., 2016; Kant et al., 2017). Similarly, now-a-days, nanofluids have found a lot of importance in PV cooling due to their higher thermal conductivity than base fluid, providing better heat transfer characteristics (Mota and Regitano, 2012; Elsheikh et al., 2018; Devendiran and Amirtham, 2016).

This review article covers almost all recent advancements in efficiency enhancements of photovoltaic through cooling with PCM as well as nanofluids which are highly efficient & renowned energy storage media now-a-days as mentioned above. In first section of this review paper, presented are the experimental & numerical investigations performed by researchers (Arteconi et al., 2017; Japs et al., 2013; Gkouskos and Tsoutsos, 2012; Hasan et al., 2014; Lo Brano et al., 2014; Mahamudul et al., 2016; Lo et al., 2013; Stropnik and Stritih, 2016; Hasan et al., 2017; Waqas and Ji, 2017; Emam et al., 2017; Kant et al., 2016; Groulx and Biwole, 2014; Sarwar et al., 2003; Ceylan et al., 2017; Hasan et al., 2007; Khanna et al., 2017; Ciulla et al., 2012; Tanuwijava et al., 2013; Nižetić et al., 2018; Ho et al., 2015; Hachem et al., 2017; Arnaud et al., 2010; Elarga et al., 2016; Klugmann-Radziemska and Wcisło-Kucharek, 2017; Machniewicz et al., 2015; Hasan et al., 2010; Tan et al., 2017; Huang et al., 2011; Huang et al., 2004; Sharma et al., 2015; Sharma et al., 2016; Park et al., 2014; Kibria et al., 2016; Ho et al., 2012; Ho et al., 2013; Sharma et al., 2017; Cellura et al., 2008; Japs et al., 2016; Yalçin and Öztürk, 2013; Smith et al., 2014; Indartono et al., 2016; Waqas et al., 2017;9.; Atkin and Farid, 2015; Hassan et al., 2014; Hasan et al., 2014; Hasan et al., 2016; Hasan et al., 2015; Biwole et al., 2013; Nehari et al., 2016; Nehari et al., 2016; Huang et al., 2006; Klemm et al., 2017; Maiti et al., 2011; Luo et al., 2017; Ho et al., 2017; Jun, 2011) who employed passive approach using PCMs for photovoltaic cooling. PCM absorbed extra heat of PV-panel by using its latent heat and maintained panel at required temperature for a certain period. Some authors in this section reported that conduction within the PCM could be improved by incorporation of different kinds of fin arrangements. To overcome low thermal conductivity problem of PCM, use of composite-PCM in various studies for PV cooling is also presented. Moreover, a novel strategy adopted by a few authors, of employing two PCMs at the same time for better heat regulation, has also been reported and explained.

Second section includes studies (Malvi et al., 2011; Browne et al., 2016; Preet et al., 2017; Zhou et al., 2017; Gaur et al., 2017; Shukla et al., n.d.; Al Imam et al., 2016; Browne et al., 2016; Su et al., 2017; Browne et al., 2015; Hasan et al., 2016; Ho et al., 2014; No Title n.d.; Yang et al., 2018; Fiorentini et al., 2015; Bigaila and Athienitis, 2017; Lin et al., 2016; Fiorentini et al., 2015; Ren et al., 2017; Aelenei et al., 2014; Ceylan et al., 2016; Storage, 2010; Lin et al., 2014; Lin and Ma, 2016; Su et al., 2017; Xiang and Gan, 2015; Sudhakar et al., 2017; Veeraragavan et al., 2014; Cui et al., 2016; Qiu et al., 2015; Qiu et al., 2016; Cui et al., 2017; Veeraragavan et al., 2017) performed on hybrid PV/T systems with PCM integrated. Heat absorbed by PCM was removed by water flowing through tubes inside PCM or air near PCM. Hence, electrical and thermal energies were simultaneously produced, increasing the system overall efficiency. In third section, experimental and simulation work of many researchers (Hussien et al., n.d.; Ghadiri et al., 2015; Sardarabadi et al., 2014; Hamdan and Kardasi, 2017; Karami and Rahimi, 2014, 2016; Lari and Sahin, 2017; Al-Waeli et al., 2017; Radwan et al., 2016; Soltani et al., 2017; Hassani et al., 2016; Ebaid et al., 2018; Hasan et al., 2017; Elmir et al., 2012; Xu and Kleinstreuer, 2014; Xu and Kleinstreuer, 2014; Khanjari et al., 2017; Khanjari et al., 2016; Paper, 2017; Sdqho et al., n.d.; Sathieshkumar, 2017; Michael and Iniyan, 2015; Gangadevi et al., 2013; Sardarabadi et al., 2017; Chandrasekar et al., 2013; Hussien and Noman, 2015; Al-Shamani et al., 2016; Karami and Rahimi, 2014; Rejeb et al., 2016; Lelea et al., 2015; Jin and Jing, 2017; An et al., 2016; Hassani et al., 2016; Jing et al., 2015; Otanicar et al., 2013;5.; Taylor and Otanicar, 2015; Cui and Zhu, 2012; Saroha et al., 2015; Taylor et al., 2012; Crisostomo et al., 2017; Taylor et al., 2013; An et al., 2016; DeJarnette et al., 2015; DeJarnette et al., 2014; Hjerrild et al., 2016; Al-Waeli et al., 2017) on hybrid PV/T systems using nanofluids, is presented. Nanofluids can be used both as coolant for PV cooling due to better heat transfer characteristics than base fluid and as optical filter for selective absorption of irradiations falling on PV-surface. This review compiles most of the reported studies on PV temperature regulation and

efficiency enhancement of PCM based PVT system both in electrical output and thermal energy utilization. First in Section 2, a detailed review is presented for PV used with only PCM for thermal regulation and then various PCM based PVT systems (using air, water and other fluids) are discussed. Sections 3 and 4 further present advances in PVT systems using simple nanofluids and combination of nanofluids with PCMs respectively. Few reported studies indicate that use of PVT systems combined with nanofluids and PCMs can improve the uniform temperature regulation with reduction in PV temperature, also better thermal and electrical efficiency can be achieved when compared to simple PCM based PVT system. However, there are still various questions to be looked into including the high cost, PCM leakage and nanofluids agglomeration etc. in order to come up with a sustainable solution.

Fig. 1 shows a schematic approach that how different classifications of PV systems thermally managed with PCMs and nanofluids, are made in this article.

2. PV/PCM system

PV modules can be cooled using passive approach with phase-change materials. When PCM is incorporated at rear surface of PV, it can absorb extra heat of PV-panel in the form of latent heat and so, can maintain the panel surface at required temperature for a certain period. Heat of panel is transferred inside PCM by conduction which can be enhanced and regulated better by incorporation of different kinds of fins arrangements. To overcome low thermal conductivity problem of PCM, use of composite PCM is a good approach. Moreover, a novel strategy of employing two PCMs at the same time for better heat regulation, can be implemented for round-the-year performance and it is explained at the end of this section.

Fill factor (FF) of photovoltaic panel is given as:

$$FF = \frac{P_m}{V_{OC} * I_{SC}}$$

where, maximum electrical power output "P_m" is determined as:

$$P_m = V_m * I_m$$

Now, electrical efficiency "η_{el}" can be calculated as:

$$\eta_{el} = \frac{P_m}{A_c G} = \frac{V_m * I_m}{A_c G} = \frac{V_{OC} * I_{SC} * FF}{A_c G}$$

2.1. PV with pure-PCM

Arteconi et al. (2017) investigated that use of thermal energy storage material like PCM could recover electricity loss of PV occurring from high operating temperature. Japs et al. (2013) reported effect of using PCM for PV-panels cooling while comparing it with conventional-PV. Panel temperature reduced by 7 °C with PCM and showed lesser fluctuation in temperature, giving more uniformity. Gkouskos and Tsoutsos (2012) found effectiveness of PCM in cooling of PV-panels in Mediterranean regions like Greece. PCM when integrated with PV-module, maintained cell temperature close to about 25 °C which is ideal for PV.

Effect of major thermo-physical properties of PCM on PV-temperature regulation was reported by Hasan et al. (2014). 3 PCM classes (paraffin wax, fatty acids-mixture & salt-hydrates) were examined thoroughly using DSC & THM. High thermal conductivity, high latent heat, reasonable melting & solidification range, less under cooling etc. had prominent effect on PV-temperature regulation. Lo Brano et al. (2014) developed simple numerical model of PV-PCM system using finite difference method and compared it with experimental system using RT27 for model validation. Proposed model was found reliable as measured and calculated Si-temperature differed by only 7% on average. Mahamudul et al. (2016) regulated PV-module temperature with RT35 as PCM. Cell temperature dropped by 35 °C in simulation and by 10 °C in experiment carried out on sunny day with 2 cm PCM layer thickness. PCM latent heat kept temperature (35 °C) constant for more than 5 h. Fig. 2 shows how experimental prototype was developed.

Lo et al. (2013) found good agreement between results of numerical

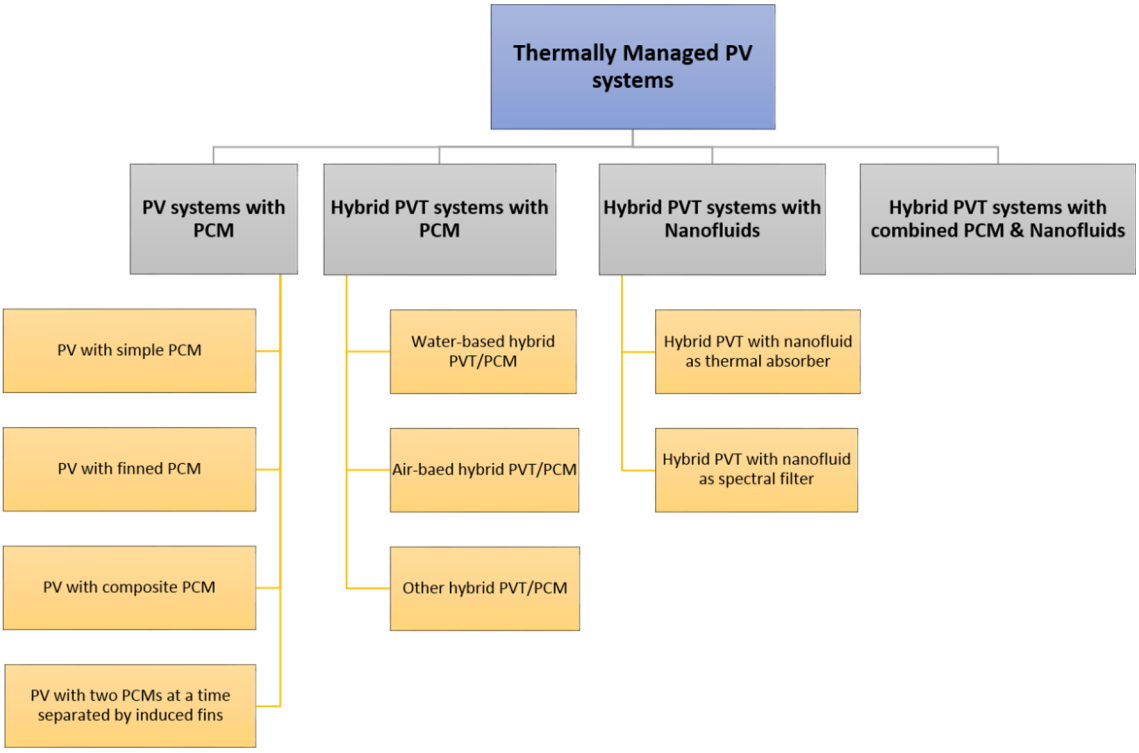


Fig. 1. Classifications of PV systems thermally managed with PCM and nanofluids.

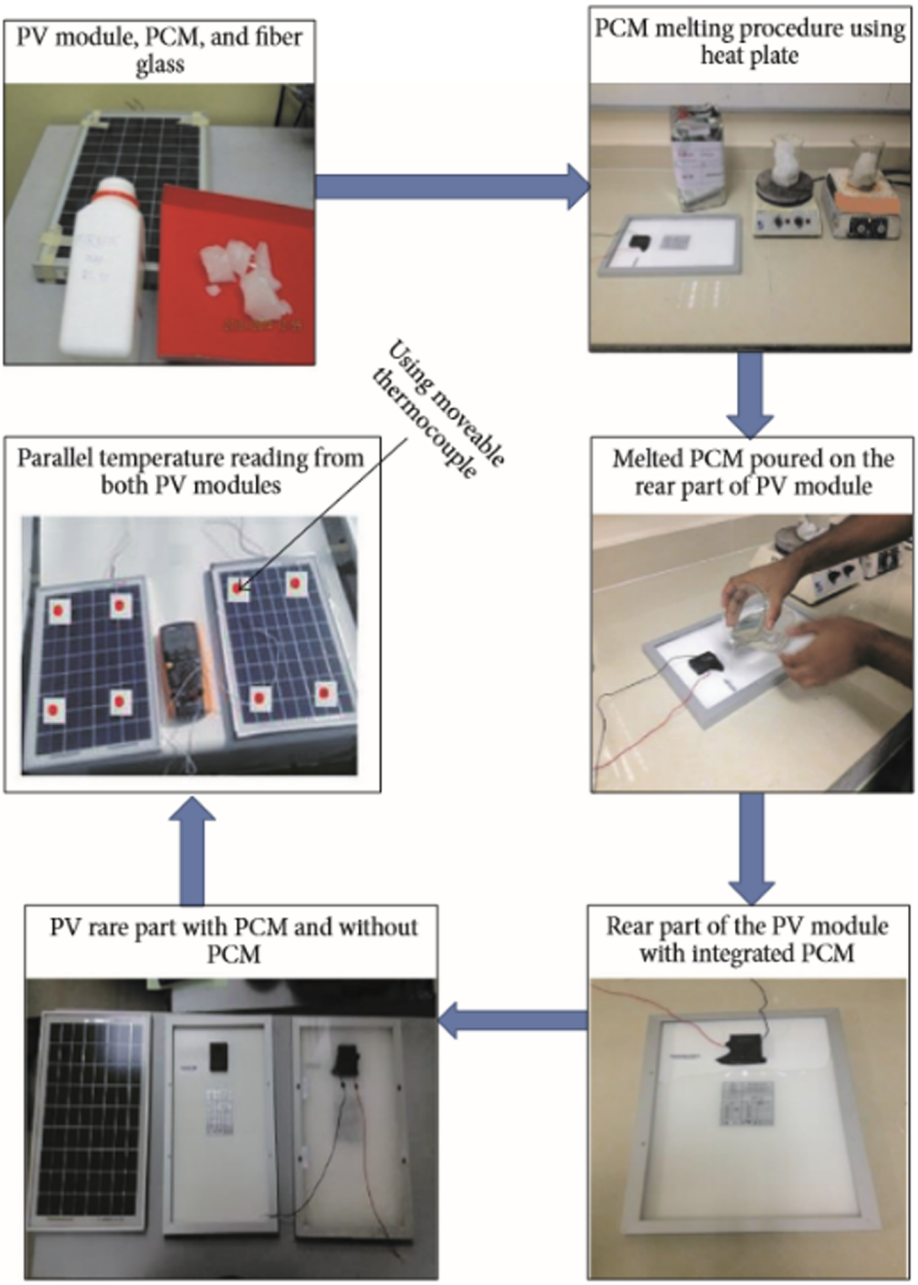


Fig. 2. Development of experimental prototype (Mahamudul et al., 2016).

simulation and experimental work for PV-PCM system. Proposed thermal model was found reliable for both sunny & cloudy days as average difference between both results was less than 5%. Stropnik and Stritih (2016) investigated PV-PCM system using RT28HC under cloudy weather of Ljubljana, a maximum of 35.6 °C while an average 14.4 °C lowering of PV-temperature was observed. Reason was effective thermal management by PCM due to its high latent heat (245 kJ/kg) as PV-temperature remained nearly constant during phase-change from 12:30 to 14:00. Simulation results also stated 7.3% increase in electrical power for a complete year. Hasan et al. (2017) used paraffin wax RT 42 for cooling of PV-panels in hot climate of UAE. During peak hot & cool months, PCM couldn't perform effectively due to incomplete solidification in night & partial melting in day-time, respectively. Maximum drop in cell temperature of 13 °C was determined in April. Power output enhanced by 5.9% in complete year showing effectiveness of PCM for PV cooling. Waqas and Ji (2017) found effectiveness of using PCM-filled moveable shutters on PV-panel back. Shutters opened during night, thus providing solution for PCM's incomplete solidification problem at night. PCMs with T_m = 35 °C & T_m = 30 °C showed maximum efficiency in June & January respectively, when system was tested under climate of Islamabad, Pakistan. Emam et al. (2017) explained effects of inclination angle (−45°; 0 °, 45°, 90°), C.R (5, 20) & PCM ($CaCl_2{\cdot}6H_2O$) layer thickness (50, 200 mm) on PV-efficiency & temperature uniformity. At inclination angle of 45°, PV-efficiency was maximum (17%) due to lowest average cell temperature. Temperature uniformity was maximum at 90^0 while worst at −45°. Melting time of PCM decreased by increasing C.R and decreasing PCM layer thickness. (Fig. 3)

Kant et al. (2016) explained through CFD analysis, that heat transfer was enhanced by convective mode during melting stage of PCM. Cell temperature dropped by 6 °C & 3 °C with convection & conduction heat transfer, respectively showing convection dominance. Increasing tilt angle increased convection inside PCM while increasing wind velocity enhanced convective heat losses, thus improving PV-efficiency. Another

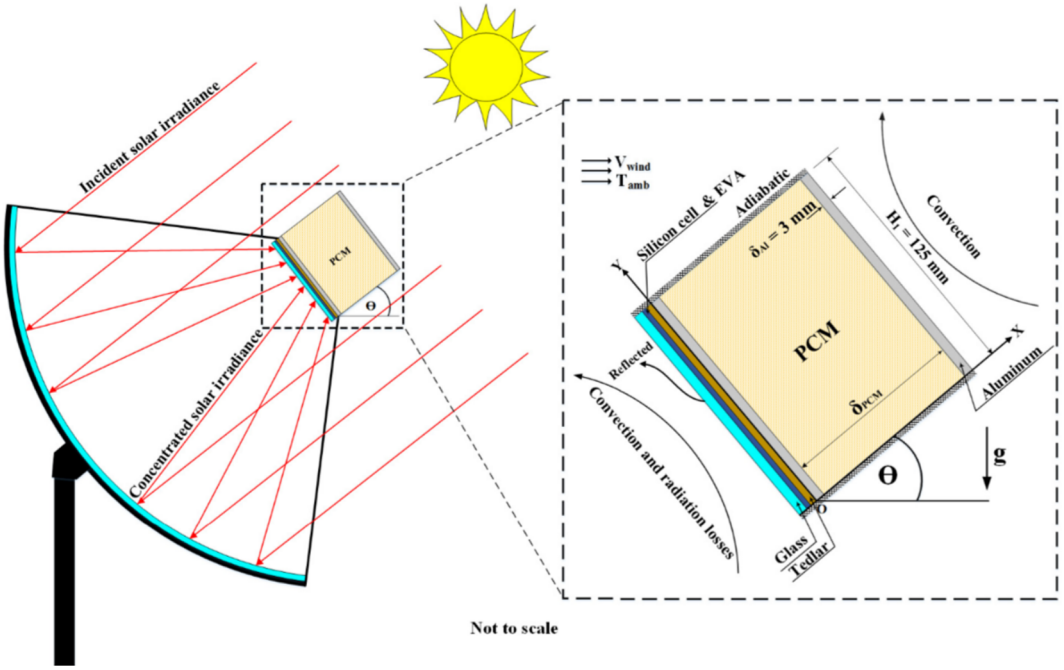


Fig. 3. Schematic diagram of concentrated PV system (Emam et al., 2017).

numerical study was done on investigation of PV/PCM system by varying inclination angle from 0 to 90^0 by Groulx and Biwole (Groulx and Biwole, 2014). At 0 °C, heat transfer was purely by natural convection while at 90^0, only conduction occurred. Hence, performance increased with increasing inclination angle. About 23 °C more temperature lowering was observed in vertical position than in horizontal position. Sarwar et al. (2003) investigated effect of varying solar irradiations (i.e. 1000, 750, 500 W/m^2) using solar simulator on cell temperature of PV-module which contained PCM (RT20) in Al container at its back. Heat absorption in latent heat of PCM was higher at higher irradiance as maximum drop in temperature was obtained in case of 1000 W/m^2 followed by 750 & 500 W/m^2.

Ceylan et al. (Ceylan et al., 2017) reported through mathematical modeling that paraffin wax was very effective in reducing cell temperature. At C.R = 3, temperature was above 100 °C for panel without PCM, while it was between 80 & 100 °C for panel with PCM (T_m = 47 °C). Hasan et al. (2007) found that a PCM used for PV cooling at one geographical location, might not necessarily perform the same way at another geographical location. Moreover, thermal conductivity of both PCM itself as well as container containing PCM at PV-panel rear, should be high for best performance (Khanna et al., 2017) used RT25HC for passive cooling of PV and evaluated its performance by numerical study. Tilt angle of PV-panel was varied from 0^0 to 90^0 with PV-PCM system performing best at 90^0 through increasing convection inside PCM with tilt angle. Peak reduction (19 °C) in PV-temperature was achieved using PCM while power & efficiency enhanced by 11.1 & 11.8%. Ciulla et al. (2012) performed numerical simulation of PV/PCM system using summer conditions of Italy and validated it with experimental results. For sunny period, good agreement was found between simulation and experimental results with mean absolute error of just 0.61 °C while error was significant at night due to improper estimation of radiations. Tanuwijava et al. (Tanuwijava et al., 2013) numerically investigated effectiveness of using MEPCM with PV-modules for efficiency enhancement by varying A.R (0.277 & 1) and PCM m.p (26 & 34 °C). PV-cell efficiency enhanced with A.R = 0.277 while reduced with A.R = 1. Average efficiency increased equally with both PCMs. Nižetić et al. (2018) evaluated performance of pork fat as PCM by comparing it with conventional organic PCM for PV cooling. By modelling both systems for a year using climate of Croatia, negligible performance difference was found due to quite similar physical properties i.e. m.p range & latent heat of fusion. Pork fat was environment friendly and had lower initial cost, but it wasn't economically viable due to its degradation in thermo-physical properties on long-term basis.

Ho et al. (2015) investigated through numerical simulations, the effect of PCM melting point (28 & 30 °C) and its thickness (3 & 5 cm) on PV-temperature regulation. With 30 °C PCM, highest PV-temperature occurred with delay as compared to 30 °C PCM. Amount of heat absorbed by 3 cm PCM thickness was lesser than 5 cm, so it resulted in lower reduction in PV-temperature. Power output increased by maximum 2.1% with 30 °C PCM of 5 cm thickness. Hachem et al. (2017) used white petroleum-jelly for PV passive cooling by attaching it to PV-panel back. 3% efficiency enhancement while 2.7 °C average lowering of PV-temperature was found. Arnaud et al. (2010) did experimental investigation by artificial sunlight followed by numerical simulation of PV-panels coupled with PCMs having melting points 27 °C and 45 °C respectively. For 27 °C PCM, constant temperature phase-change process occurred for longer duration and temperature drop was higher as compared to 45 °C PCM. Power output enhanced by 15 to 25% with 27 °C PCM. Elarga et al. (2016) compared the performance of DSF + PCM-PV layer with DSF + PV layer through numerical simulation. RT42 was used for climatic conditions of Venice and Helsinki while for warmer climate of Abu Dhabi, RT55 was used. PCM thermally managed PV-module by temp. lowering of max. 10, 20 & 17 °C, resulting in power output enhancement of 5, 8 & 5% for Venice, Helsinki & Abu Dhabi respectively, with DSF + PCM-PV layer as compared to DSF + PV layer.

Radziemska & Kucharek (Klugmann-Radziemska and Wcisło-Kucharek, 2017) studied heat flow in three PCMs i.e. Paraffin 42–44, RT 22, Ceresin having melting points in range of 42–72 °C, 20–23 °C, 61–78 °C respectively. RT 22 converted into liquid very early while Ceresin remained in solid form, so Paraffin 42–44 with 20 mm thickness was selected for thermal management of PV using solar simulator. Cell temperature lowered by 7 °C, was maintained for 5 h. Machniewicz et al. (2015) stated through results of simulation carried out for 4 different PCMs used for PV cooling, that with low transition temperature, PCM melted constantly, while PV-temperature remained very high with high transition temperature. Conclusion was that with transition temperature range of 18 to 25 °C, PCM was most effective in lowering PV-

temperature. Hasan et al. (2010) experimentally tested performance of 4 different PV/PCM systems. System A & C had Al container with internal width as 5 & 3 cm respectively while system B & D had Perspex container with internal width of 5 & 3 cm respectively. System A performed best among other systems with 10 °C drop in temperature, maintained for 5 h at 1000 W/m^2. Higher thermal conductivity of Al than Perspex provided better heat transfer from PV to PCM while more internal width of container allowed for more latent heat storage inside the PCM. Tan et al. (2017) determined effectiveness of PCM (RT27) under outdoor conditions as cell temperature dropped by maximum 15 °C while efficiency improved by 5.39% as compared to system without PCM. Huang et al. (2011) reported thermal regulation of PV/PCM system through indoor experiment. For best performance, convection inside PCM played an important role which could be increased by fins presence. PCM (RT27) passively reduced PV-temperature from 63 °C to 42 °C showing its effectiveness in PV cooling.

Huang et al. (2004) stated through numerical simulation and experimental work results that with 3 cm PCM depth, PV-temperature remained under 35 °C. PCM presence reduced temperature to great extent due to natural convection flows within molten PCM. Sharma et al. (2015) developed an indoor BICPV using compound parabolic concentrator and performed experiment using solar simulator (1000 W/m^2). PCM heat sink was attached under PV-module. PCM (RT42) produced temperature uniformity in module avoiding the hot spots and lowered PV-temperature by 5.2 °C while efficiency enhanced by 15.9%. Sharma et al. (2016) performed 2 h-indoor experiment for PV cooling through PCM (RT42) with the help of solar simulator. At 1000 W/m^2 irradiance, PV-temperature reduced by maximum 12 °C. Moreover, power & efficiency also enhanced by 16.67% & 7.7%, respectively. PV-efficiency increased with increasing irradiance, reached maximum at 1000 W/m^2 and then decreased a little at 1200 W/m^2. Park et al. (2014) collected 1 month data to compare performance of PV-PCM system with conventional-PV at Song-do, South Korea. Maximum 5 °C drop in PV-temperature was achieved by PCM (T_m = 25 °C). Power output increased from 1.9 to peak 1.96 W, while PV-efficiency enhanced from 8.06 to 8.34%. Efficiency increased very little by increasing PCM thickness from 30 to 50 mm, while optimum melting point of PCM was 25 °C. Fig. 4 shows schematic diagram of PV/PCM module.

BIPV performance was numerically tested by Kibria et al. (2016) through incorporating 3 PCMs (RT20HC, RT25HC, RT28HC) separately for performance comparison. RT28HC showed best performance due to its highest latent heat, as cell temperature reduced by 18 °C while efficiency enhanced by 5%. Ho et al. (2012) did numerical simulation of BIPV integrated with MEPCM layer, using winter and summer conditions. Effects of A.R (0.277 & 1.0) and T_m (26 & 34 °C) of MEPCM on BIPV performance were investigated. A.R = 0.277 increased efficiency while 1.0 decreased efficiency. With 26 °C MEPCM & A.R = 0.277, efficiency increased by 0.42 & 0.13% in winter & summer, respectively.

Effect of layer thickness (2 & 3 cm) & m.p (30 & 32 °C) of MEPCM on BIPV performance, was well examined by Ho et al. (2013) through CFD numerical simulation. Results showed that best performance was achieved with 3 cm-MEPCM layer having T_m = 30 °C. Sharma et al. (2017) compared performances of naturally-cooled & PCM-cooled photovoltaic systems. Average drop in PV-temperature of 9.6 °C was obtained using PCM. Cellura et al. (2008) performed numerical simulation of PV-module with paraffin wax as PCM (T_m = 28–32 °C) coupled at its back in summer condition. Results showed temperature reduction of about 19 °C with efficiency increase from 12 to 26%, resulting in reduced CO_2 emissions. Japs et al. (2016) stated through experimental investigation that although PCM reduced cell temperature but when seen from technical & economical viewpoint, performance difference with ref-PV, was insignificant. Economics could be improved by graphite addition in PCM. Yalçin and Öztürk (2013) determined feasibility of using PCMs for increase in PV energy output. PCM could kept PV-temperature constant for 2 to 3 h a day. But cost-analysis revealed that PV/PCM systems can give annual extra electrical yield of just 1 to 3% and so was not economically viable although BIPV/PCM systems were comparatively more viable. Smith et al. (2014) performed numerical simulation of PV/PCM system, globally over grid of 1.5^0 * 1.5^0. For all locations, T_m of PCM was varied from 0 to 50 °C to determine optimum T_m. Regions with higher T_{amb} required PCM with higher T_m. PCM inclusion was more effective in higher insolation regions than the lower ones. Energy output was more than 6% for eastern Africa & Mexico, 2 to 5% for Europe & about 5% for regions like South America, Southern Asia, Arabia, Indonesian archipelago. Cost-analysis revealed ineffectiveness of using PCMs for performance enhancement of PV.

Table 1. shows summary of work proposed by different researchers for PV cooling using pure-PCM.

2.2. Discussion

PCM is quite effective in absorbing PV-panel extra heat which is not converted into electrical energy. Latent heat of PCM helps in lowering panel temperature, bringing it closer to 25 °C and maintaining it almost constant during peak sunny hours. Maximum temperature reduction of about 35 °C is brought using PCM by (Mahamudul et al., 2016); (Stropnik and Stritih, 2016); (Waqas and Ji, 2017). Most of the work is done in summer conditions due to high ambient temperature & solar intensity. In summer, PCM mostly used is RT42 due to high PV-surface temperatures (as seen in Fig. 5 (top)), while PCM with melting point in range of 25-30 °C is mostly used in winter conditions (as seen in Fig. 5 (bottom)) as less reduction in PV-temperature is required.

As seen from Fig. 6, PV-efficiency enhancement up to 16% can be achieved by PCM. Comparison of both graphs shows that researchers found more efficiency enhancement in summer than in winter, showing that PCM incorporation is more effective in summer. Reason behind this is higher heat absorption in summer by PCM in the form of its latent heat.

From Fig. 7, it is found that significant increase in PV power output can be achieved with PCM as compared to without PCM. Max. enhancement is 23% by (Stropnik and Stritih, 2016) who used RT28 HC which brought panel temperature close to 25 °C, where PV shows maximum power output.

2.3. PV with finned-PCM

Indartono et al. (2016) used yellow petroleum jelly as PCM and inserted it into Al tubes for passive cooling of PV-panels, placed in 2 configurations ('PV on stand' & 'PV on roof'), for which average PV-temperature dropped by 0.4 & 2.7 °C respectively. Similarly, average increase in efficiency was 6 & 21.2% respectively. Former configuration

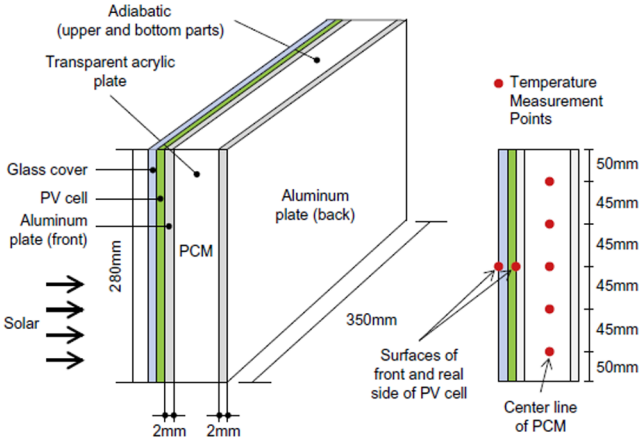


Fig. 4. PV/PCM module – schematic diagram (Park et al., 2014).

Table 1
Summary of work proposed by different researchers for PV cooling using pure-PCM.

Ref no.	PV-panel specifications	Type of PV system	Nature of Work and Duration	PCM tested, melting point (^{0}C)/ Latent Heat (kJ/kg)	Peak $T_{PV\text{-}ref}$/ $T_{PV\text{-}PCM}$	Key Findings
(Japs et al., 2013)	30 W monocrystalline PV-panels	PCM filled in bags attached to back of PV-panel	Outdoor experiment in summer at Paderborn, Germany from 15th Jul to 16th Aug	PCM, 218 kJ/kg	~54 °C/~49 °C	PCM brought uniformity in panel temperature and peak reduction was 7 °C
(Mahamudul et al., 2016)	PV-module	PV-module with PCM integrated at its back	Numerical Simulation	RT35, 35 °C	70 °C/35 °C	PV-PCM system showed 35 °C temperature drop as compared to PV system.
(Stropnik and Stritih, 2016)	250 W CS6P-M PV-panel with dimensions 340 × 280 (mm)	PV with PCM box attached on behind and enclosed with acrylic glass	Outdoor experiment at Ljubljana, Slovenia, from 14th to 21th October 2013	RT28 HC, 28 °C/245 kJ/kg	72 °C/44 °C	PV-surface temperature reduced by 35.6 °C.Electrical efficiency enhanced by 23.2%.
(Hasan et al., 2017)	40 W polycrystalline EVA encapsulated panels, 53 × 63(cm)	PV with PCM container attached at behind	Outdoor experiment at Al-Ain, UAE for a complete year 2015	RT42, 38–43 °C/145 ± 7% kJ/ kg	–	PV-cell temperature dropped by maximum 13 °C in April. Average increase in electrical power output was 5.9% for complete year.
(Waqas and Ji, 2017)	Conventional-PV Panels with PCM filled shutters on back	PV with PCM-filled shutters on back	Numerical simulation using conditions of 1st week-Jan & 2nd week-Jun in Islmbd, Pakistan	RT44, 44 °C (peak)/250 kJ/kg	90 °C/55 °C (2nd week-Jun) 53 °C/43 °C (1st week-Jan)	In June, PCM with T_m = 35 °C gave maximum efficiency as PV temp. reduced by 35 °C while in January, PCM with T_m = 30 °C was found most efficient with 17 °C drop in temp.
(Kant et al., 2016)	PV-module	PV-module coupled with PCM at its back	Numerical simulation with climatic conditions of May in City of Allahabad, India	RT35, 35 °C/240 kJ/kg	60 °C/54.9 °C	During melting stage of PCM, convective mode enhanced heat transfer more than by conduction (i.e. 6 °C). Moreover, increasing the tilt angle also increased convection.
(Sarwar et al., 2003)	Conventional-PV-panel	PV using PCM in aluminum container at back	Indoor experiment using Solar simulator (500, 750, 1000 W/ m^2) for 6 h and a half	RT20, 20 °C	~57 °C/~56 °C (1000 W/m^2) - ~50 °C/~48 °C (750 W/m^2) - ~43 °C/~42 °C (500 W/m^2)	By increasing solar irradiations, cell temp dropped more due to PCM (i.e. 8 °C, 9 °C and 13 °C for 500, 750 and 1000 W/m^2 respectively).
(Khanna et al., 2017)	PV-module	PV with PCM box attached at behind	Numerical simulation for 3 h	RT25 HC, 26.6 °C/232 kJ/kg	~54 °C/~40 °C	Increasing tilt angle from 0^0 to 90 °C increased convection rate in PCM. Temp. reduced by 19 °C while efficiency enhanced by 11.8%.
(Tanuwijava et al., 2013)	PV-module	PV-module with MEPCM attached at back	Numerical simulation	MEPCM, 26 °C & 34 °C/172 kJ/ kg	~41.25 °C/ ~40.4 °C	Both PCMs (T_m = 26 °C & 34 °C) showed almost equal increase in average efficiency. Efficiency enhanced with 0.277 AR while reduced with 1.
(Ho et al., 2015)	PV-module	MEPCM-PV module floating on water surface	Numerical simulation with 2-day summer conditions of Taiwan	PCM_1: 30 °C PCM_2: 28 °C	31.8 °C (PCM_1) 32.3 °C (PCM_2)	5 mm thickness of PCM layer resulted in best performance.
(Hachem et al., 2017)	Two 30 W polycrystalline PV-panels, 50 × 50 (cm)	Prototype 1: conv. PV panel - Prototype 2: PV with pure-PCM at back	Outdoor experiment at Al-Khyara, West Bekaa of Lebanon on 16th May 2015	PCM (white petroleum jelly), 36–60 °C, k = 0.18 W/mK	66.3 °C/65.6 °C	PV-efficiency enhanced by 3% while average surface temp. lowered by 2.7 °C.
(Elarga et al., 2016)	140 W Si PV-module	PV and PCM in double skin facade	Numerical Simulation	RT42, 41 °C, 175 kJ/kg (for Venice/Helsinki) - RT55, 55 °C, 170 kJ/kg (for Abu Dhabi)	~68 °C/~46 °C (Venice) - ~84 °C/~54 °C (Helsinki) - ~84 °C/~66 °C (Abu Dhabi)	PV temp. reduced by 10 °C, 20 °C and 17 °C, enhancing power output by 5%, 8% and 5% for Venice, Helsinki and Abu Dhabi respectively.
(Klugmann-Radziemska and Wcisło-Kucharek, 2017)	10 W PV-panel with dimensions 340 × 280 (mm)	PV with PCM box attached at behind	Indoor experiment using Solar Simulator (1000 W/m^2) for 4.5 hrs	Paraffin 42–44, 42–72 °C/ 130 kJ/kg	72 °C/64 °C	Max. reduction in cell temp. of 7 °C was maintained for 5 h. 20 mm PCM thickness resulted in best performance.
(Sharma et al., 2015)	8 LGBC crystalline Si cells (116 × 6 mm) soldered in series	BICPV	Indoor experiment (2 hrs) using Wacom Solar Simulator (1000 W/m^2)	RT42, 41 °C/145 kJ/kg	60.9 °C/58.7 °C	PCM brought temp. uniformity in module avoiding the hot spots and lowered the module temp. by 5.2 °C while electrical efficiency enhanced by 15.9%.

(*continued on next page*)

Table 1 (*continued*)

Ref no.	PV-panel specifications	Type of PV system	Nature of Work and Duration	PCM tested, melting point (°C)/ Latent Heat (kJ/kg)	Peak $T_{PV\text{-}ref}$/ $T_{PV\text{-}PCM}$	Key Findings
(Sharma et al., 2016)	5 LGBC crystalline Si cells (116 mm × 6 mm) connected in series	PCM containment integrated with BICPV-module	Indoor experiment with Wacom Solar Simulator (1000 W/m^2) for 2 h	RT 42, 38–43 °C/174 kJ/kg	55.2 °C/48.8 °C	PV-surface temp. reduced by max. 12 °C. Electrical power and efficiency enhanced by 16.67% and 7.7% respectively.
(Park et al., 2014)	Two 10.9 W monocrystalline panels, 350 × 280 (mm)	PV/PCM module installed on a vertical wall surface	Outdoor experiment at Songdo, Incheon, South Korea, from 17th May to 17th June 2012	25 °C/184 kJ/kg	36 °C/-32.5 °C	Cell temp. lowered by 5 °C. Optimum T_m of PCM was 25 °C. Increase in PCM thickness from 30 to 50 mm had little effect on efficiency.
(Kibria et al., 2016)	PV-module	PV with PCM incorporated at its back	Numerical Simulation	RT20HC, 21.23 °C/140 kJ/kg - RT25, 25 °C/148 kJ/kg - RT28HC, 28 °C/245 kJ/kg	–	RT28 HC due to its high latent heat than RT20 HC & RT25, performed best reducing PV temp. by 18 °C while enhancing efficiency by 5%.
(Ho et al., 2012)	PV-module	BIPV integrated with MEPCM layer	Numerical Simulation using winter and summer conditions	MEPCM, 26 °C & 34 °C	–	Both MEPCMs showed insignificant difference in efficiency enhancement. Efficiency increased by 0.277 AR while decreased by 1.0 AR.
(Ho et al., 2013)	PV-module	BIPV integrated with water-saturated MEPCM layer	Numerical Simulation	MEPCM, 30 °C & 32 °C	38.5 °C/34.1 °C (3 cm)	Best performance of BIPV was achieved with MEPCM of T_m = 30 °C and 3 cm layer thickness.
(Cellura et al., 2008)	PV-module	PV coupled with PCM at its back	Numerical simulation using conditions of typical summer day in Palermo, Italy	RT42, 28–32 °C/157–179 kJ/kg	-83 °C/-64 °C	Cell temp. reduced by 19 °C, so efficiency increased from 12% to 26%.

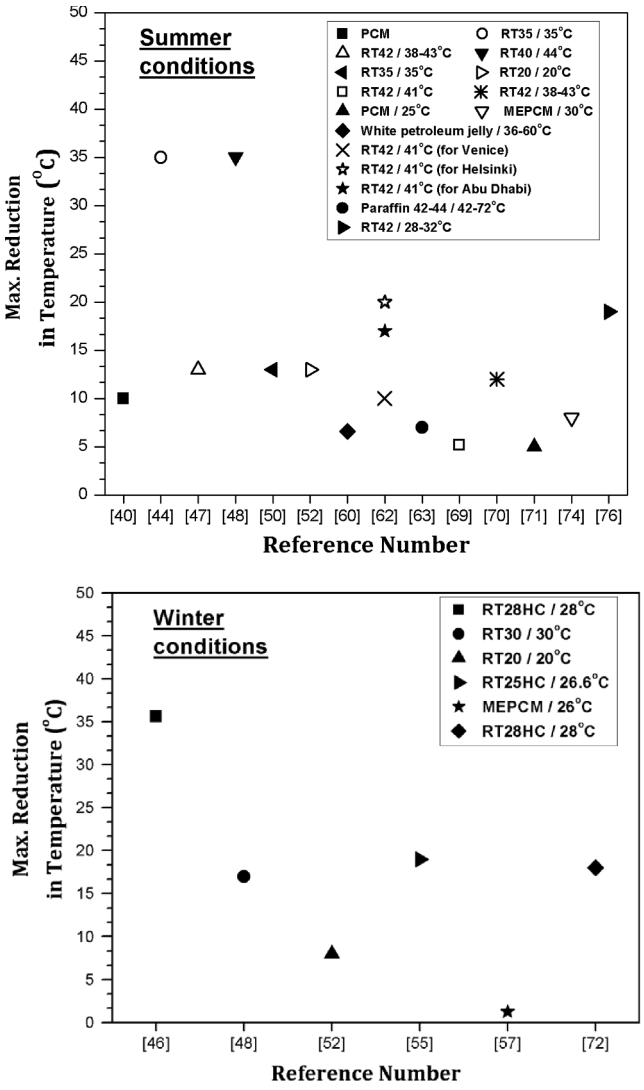


Fig. 5. Maximum reduction in PV-temperature using PCM as compared to conventional-PV achieved by different researchers in summer (top) & winter (bottom) conditions.

was more effective as wind generated convection on back of PV-surface, further cooling it. A novel strategy of using PCM for passive PV cooling was used by Waqas et al. (2017). PCM (RT24) was filled in 11 Cu tubes of 14 mm diameter attached to PV-panel back with thermal grease. Presence of metallic tube enhanced convective heat transfer leading to better control on temperature rise. This resulted in maximum temperature drop of 15 °C while PV-efficiency enhanced by 3%. Huang et al. (2004) performed numerical simulations of 3 systems i.e. conventional-PV, PV/PCM without fins and with 2 internal fins. For 3rd system, PV-temperature remained constant at 36.4 °C for 80 min. Fins increased temperature uniformity on panel surface and reduced its temperature to great extent but due to limitation in movement of molten PCM, only certain number of fins could be inserted (Fig. 8).

Atkin & Farid (Atkin and Farid, 2015) studied effectiveness of using fins inside PCM for thermal management of PV-panels. Presence of fins inside PCM increased thermal effects resulting in enhanced PV cooling as compared to system without fins. Sharma et al. (2017) compared performances of naturally cooled, PCM-cooled with & without micro-fins PV-panels, respectively. Presence of PCM (RT42) reduced PV-temperature by 9.6 °C but fins inclusion reduced it further by 1.1 °C due to enhanced conduction rate by PCM contact with metal. Similarly, power output increase with & without fins, was 5.35 & 4.8%

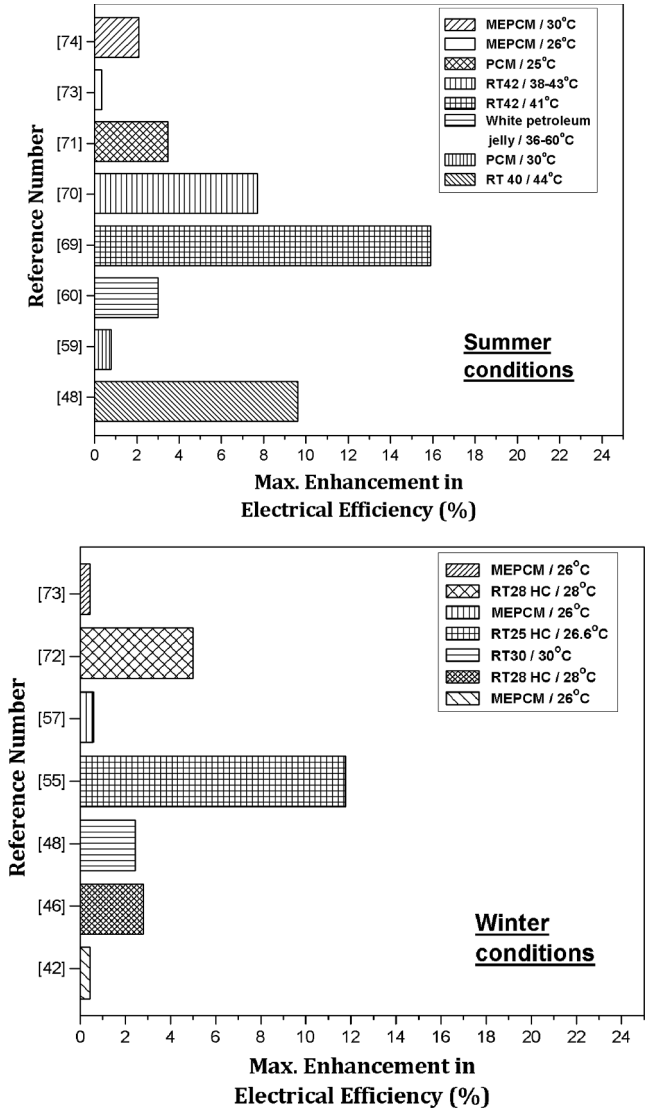


Fig. 6. Maximum enhancement in PV electrical efficiency using PCM as compared to conventional-PV achieved by different researchers in summer (top) and winter (bottom) conditions.

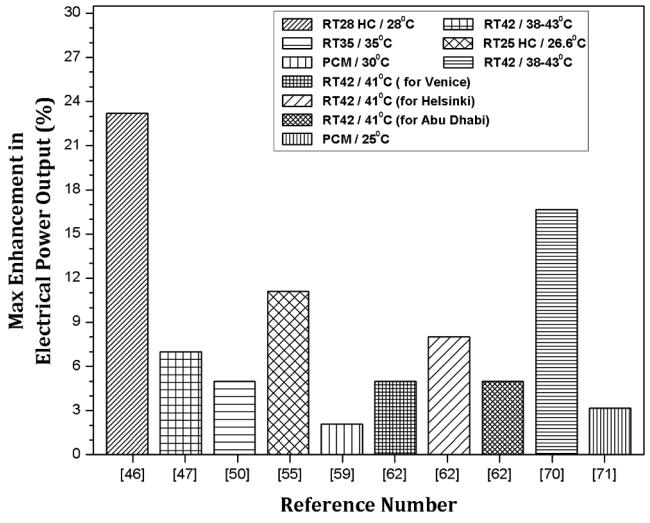


Fig. 7. Maximum enhancement in PV power output using PCM as compared to conventional-PV achieved by different researchers.

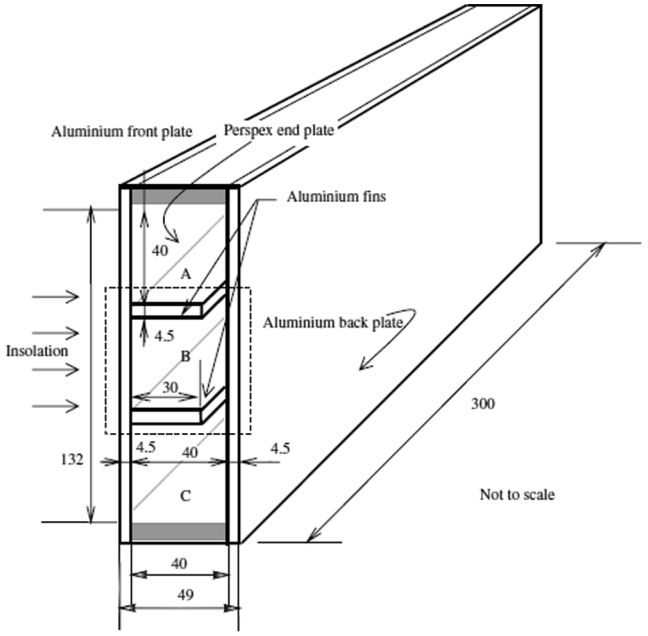


Fig. 8. Experimental PV/PCM system internally incorporated with fins – Schematic diagram (mm) (Huang et al., 2004).

respectively. Huang et al. (2011) reported effect of varying fin spacing of (0, 4, 8, 12, 24, 33 mm) on thermal regulation behavior of PV/PCM system. Although fins presence enhanced convection inside PCM (RT27) but time required for complete PCM melting decreased with increasing number of fins. Maximum drop (i.e. 10 °C) in PV-temperature as compared to PV/PCM without fins, was observed in case of 4 mm fin-spacing due to maximum contact of metal surface with PCM. Hassan et al. (2014) experimentally examined PV/PCM system performance in Al-Ain. In sunny days, peak temperature of ref-PV panel reached 58 °C but with finned-PCM, it reached to lower value of 47 °C. Maximum temperature reduction of 11 °C & 5 °C was observed on sunny & cloudy days, respectively.

Hasan et al. (2014) examined PV-panels incorporated with internally finned-PCM under climates of Dublin & Vehari. Eutectic of capric-palmitic acid (PCM_1) & $CaCl_2{\cdot}H_2O$ (PCM_2) were used. Peak PV-temperature reduced from 49 °C to 43 °C & 40 °C with PCM_1 & PCM_2, respectively in Dublin. While in Vehari, it dropped from 63 °C to 51 °C & 42 °C with PCM_1 & PCM_2, respectively. Results revealed higher cost-effectiveness of PV-PCM systems in Vehari than Dublin. Hasan et al. (2016) reported performance of PCM (RT42) incorporated with vertical fins on back of panel, fixed on test chamber as BIPV, at Al-Ain (Fig. 9). PCM lowered PV-temperature by 22.6 °C, enhancing power by 7.2% during daytime by utilizing its latent heat while fins presence resolved issue of PCM's low thermal conductivity.

Hasan et al. (2015) compared performances of PV-PCM systems for cooler & warmer climates of Dublin & Vehari, respectively. Capric-palmitic acid (PCM_1) & $CaCl_2{\cdot}6H_2O$ (PCM_2) with fins inserted, were used separately. PCM_2 caused more PV cooling than PCM_1 due to better thermos-physical properties. PV-PCM system was more effective in warmer than in cooler climate. Power saving by PCM_1 & PCM_2 was 4% & 11.3%, respectively in Dublin, while it was 5.1% & 13% respectively, in Vehari. Tan et al. (2017) proved effectiveness of fins inside PCM (RT27) on power output & efficiency by examining 4 different PV configurations i.e. finless, 3, 6 & 12 fins-PV/PCM under real outdoor conditions (Fig. 10). Maximum drop (15 °C) in PV-temperature was found in PV/PCM with 12 fins, with 5.39% enhancement in PV-efficiency. Reason was maximum PCM contact with metal surface leading to increased PCM melting rate.

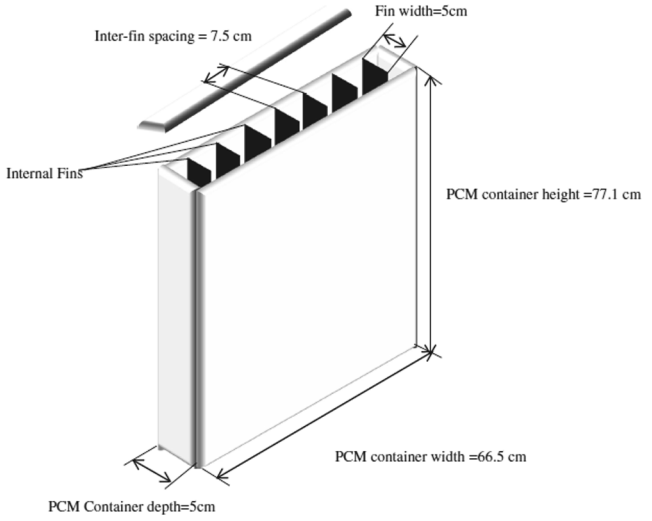


Fig. 9. PV-PCM container with fins inserted internally (Hasan et al., 2016).

Biwole et al. (2013) performed CFD analysis to explain heat & mass transfer going inside finned-PCM (RT25) attached to PV-panel back. PV-temperature without PCM reached 40 °C after just 5 min and it kept on increasing further. PCM using its latent heat, held panel temperature at 40 °C for 80 min controlling the temperature rise. Nehari et al. (2016) determined through numerical simulation the optimal fin length inside PCM (RT25) by varying fin lengths as 0, 5, 10, 15, 20, 25, 30, 35, 40 mm from panel front. Fins presence helped in even distribution of heat inside PCM, leading to enhanced PV cooling. Most effective fin lengths, helpful in hindering PV-temperature rise, were 25, 30 & 35 mm. Fins originating from panel back actually promoted temperature rise, so they were not feasible. Nehari et al. (2016) discussed effect of inclination angle on PV performance with finned-PCM (RT25) at panel back for its passive cooling. With increasing angle, conduction in PCM increased while at small angles, convection was dominant. So, at angles lower than 45^0, PCM was effective due to convection dominance. Huang et al. (2006) compared indoor performance of 3 PV-PCM systems (i.e. EA, EB, EC) with ref-PV system. EA & EB configurations had maximum 2 & 32 fins respectively, while EC contained strip Al matrix. PCM used was RT25 (MP = 26.6 °C). Although more than 30 °C lowering of cell temperature was found in all configurations, but it was higher in EB & EC than in EA, due to more metal surface area in contact with PCM. Increasing fin width maintained operating temperature constant for greater duration.

Table 2 shows summary of work proposed by different researchers for PV cooling using finned-PCM.

2.4. Discussion

As seen from Fig. 11 (top) that with use of finned-PCM, PV-temperature reduction values mostly range between 5 and 20 °C, while max. value is about 31 °C obtained by (Huang et al., 2006) who used RT25 having T_m = 26.6 °C, very close to ideal PV-temperature of 25 °C where PV-module performs at its best. Effect of fins insertion inside PCM reduced PV-surface temperature further as seen from Fig. 11 (bottom). This is due to the fact that presence of fins increased conduction heat transfer rate within PCM, which is very much required as thermal conductivity of PCM is very low. This increased transfer of excess heat from PV-surface to PCM followed by enhanced PV performance. Fins also prolonged temperature regulation period during peak hours and brought uniformity in PV-surface temperature as well. Finned-PCM system showed maximum of about 15 °C (by (Huang et al., 2011); (Atkin and Farid, 2015) more temperature drop than unfinned-PCM system.

In existing literature, maximum enhancement in PV-efficiency found is about 21% as seen from Fig. 12 (top), achieved by (Indartono et al., 2016) using yellow petroleum-jelly as PCM incorporated with fins. Maximum power enhancement is also achieved by (Indartono et al., 2016) as seen from Fig. 12 (bottom). Reason is high heat absorbing capacity of PCM along with increased conduction rate due to fins inserted inside PCM. Fins enhanced temperature regulation period during peak hours, thereby hindering power output decline. While for other researchers, power enhancement is in range of 5–12% showing effectiveness of finned-PCM.

2.5. PV with composite-PCM

Japs et al. (2013) compared PV performances with conventional & compound PCMs, having thermal conductivity values as 0.2 & 2.4 W/mK, respectively. Increase in thermal conductivity was brought at expense of PCM latent heat. PV-temperature reduced by 10 °C & 7 °C respectively, although after PCM melting, temperature of compound PCM panel rised slowly due to higher thermal conductivity. Klemm et al. (2017) introduced metallic fiber structures for enhancing thermal conductivity of PCM (RT54 HC) for better passive PV cooling. Their porosity was also varied to determine its effect on cell temperature. Presence of metallic fiber structures increased heat flow within PCM, while their porosity had negligible effect on performance. PV-temperature dropped by maximum 20 °C with 50 mm PCM thickness, which was found most appropriate after varying it from 10 to 70 mm (Fig. 13).

Maiti et al. (2011) used paraffin wax (T_m = 57 °C) with metal turnings (see Fig. 14) inside it as composite-PCM incorporated on PV-panel back. Indoor & outdoor experiments were carried out for comparison. For outdoor experiment, V-trough was placed before panel,

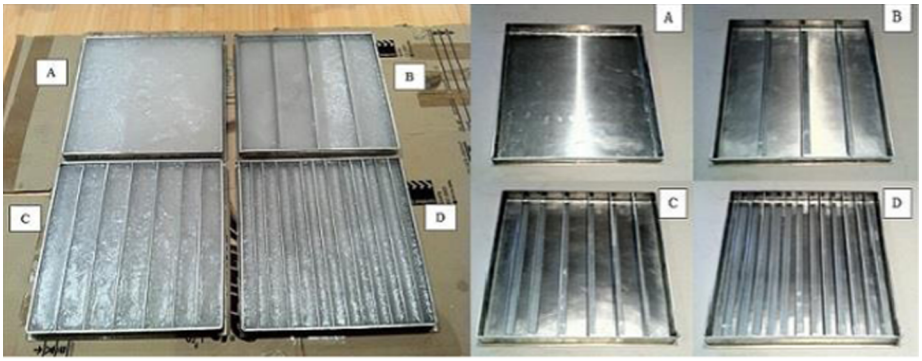


Fig. 10. PCM containments with (left) and without (right) paraffin wax in different configurations; finless (A), 3 fins (B), 6 fins (C), 12 fins (D) (Tan et al., 2017).

Table 2
Summary of work proposed by different researchers for PV cooling using finned-PCM.

Ref no.	PV-panel specifications	Type of PV system	Nature of Work & Duration	Fins Structure	PCM tested, melting point (^{0}C)/ Latent Heat (kJ/kg)	Peak $T_{PV\text{-}ref}$/$T_{PV\text{-}PCM}$	Key Findings
(Indartono et al., 2016)	Two 10 W monocrystalline panels	PCM filled in tubes on back of PV-panel with 'PV on stand & roof' configurations	Outdoor experiment at Indonesia	Aluminum rectangular tubes having cross-sections 15 × 15 (mm)	Yellow petroleum jelly, 42–55 °C/230–260 kJ/kg	PV on stand: 60 °C/ 54.3 °C - PV on roof: 44.8 °C/ 42.2 °C	"PV on stand" configuration performed better than "PV on roof" due to convection by wind.
(Waqas et al., 2017;9.)	Two 50 W mono-crystalline panels, 630 × 510 (mm)	PV with PCM filled in copper tubes attached at its back	Outdoor experiment at UST of China on May 25 & 26, 2017	11 Copper tubes of 14 mm diameter	RT24, 24 °C/250 kJ/kg	64 °C/ 59 °C	Temp. dropped by max. 15 °C. Presence of PCM in metallic tube led to better control on the temp. rise.
(Atkin and Farid, 2015)	Two mono-crystalline panels, 0.22 × 0.235 (m)	Case A: ref-PV - Case B: PV with PCM infused graphite - Case C: PV with finned heat sink - Case D: PV with infused graphite & fin heat sink	Indoor experiment using insolation provided by two 500 W halogen lamps for 24 h	Plain fin heat sink	Paraffin wax RT40 (40 °C) infused with graphite	Case B: 77 °C/75 °C Case C: 77 °C/65 °C Case D: 77 °C/66 °C	Fins inside PCM in case C & D increased thermal effects resulting in enhanced passive cooling of PV-panel.
(Sharma et al., 2017)	Conventional-PV panel	Naturally cooled, PCM cooled, BIPV with and without micro-fins	Indoor experiment using Solar simulator	20 micro-fins on the back, each having thickness of 1 mm, 5 mm height & 0.5 mm fin spacing	RT42, 38–43 °C/174 kJ/kg, (0.5 wt% CuO added)	72 °C/62 °C	Average drop in temp. was 10.7 °C & 9.6 °C with & without fins respectively while power output increased by 5.35% & 4.8%.
(Huang et al., 2011)	Conventional-PV panel	PV/PCM with fins inserted having different fin spacings	Indoor experiment using GR262 solar simulating lighting system for 4 h	Horizontal fins spacing (i.e. 4, 8, 12, 24, 33 mm)	RT27, 25–28 °C/184 kJ/kg	63 °C/44 °C	33 mm fin spacing showed max. drop in temp. (i.e. 36 °C).
(Hassan et al., 2014)	Two 30 W poly-crystalline PV-panels, 500 × 400 (mm)	PV-panel integrated with internally finned-PCM	Outdoor experiment at Al-Ain, UAE from 23/3/14 to 2/4/14	Vertical back-to-back fins with 60 mm spacing	Paraffin wax	58 °C/47 °C	With finned-PCM, PV temp. reduced by max. 11 °C & 5 °C in sunny & cloudy days respectively.
(Hasan et al., 2016)	Two 40 W polycrystalline panels, 53 × 63 (cm)	PV-PCM with fins incorporated	Outdoor experiment at Al-Ain from 23/9/15 to 7/10/15	Vertical back-to-back fins with 7 cm fin spacing	RT42, 41 °C/145 kJ/kg	68.6 °C/ 60.5 °C	Maximum reduction in PV temp. of 22.6 °C found at back surface of PV-PCM.
(Hasan et al., 2015)	Three 65 W polycrystalline EVA encapsulated panels, 771 × 665 (mm)	PV-PCM with fins inserted	Outdoor experiment at Dublin from 27th Aug to 13th Sep and at Vehari from 30th Oct to 13th Nov	Straight vertical back to back fins with 7.5 cm fin spacing	PCM_1 = capric- palmitic acid, 22.5 °C/173 kJ/kg PCM_2 = $CaCl_2{\cdot}6H_2O$, 29.8 °C/ 191 kJ/kg	At Dublin: 49 °C/43 °C (PCM_1) 49 °C/39 °C (PCM_2) At Vehari: 63 °C/46 °C (PCM_1) 63 °C/42 °C (PCM_2)	PCM_2 showed better results than PCM_1 due to better thermo-physical properties. PV-PCM system was more effective in warmer than in cooler climate.
(Tan et al., 2017)	Two multi-crystalline PV-panels (BP solar, SX 320 model)	PV/PCM with 4 configurations (finless, 3, 6, 12 fins)	Outdoor experiment for 1 h	Straight vertical fins	RT27, 27 °C/184 kJ/kg	63 °C/ 49 °C	PV/PCM with 12 fins showed best performance as PV temp. reduced by 15 °C & efficiency enhanced by 5.39%.
(Nehari et al., 2016)	PV-module	PV with finned-PCM at back	Numerical simulation for about 4 h	Horizontal fins, lengths as 5, 10, 15, 20, 25, 30, 35, 40 mm	RT25, 26.6 °C/232 kJ/kg	–	The presence of fins brought uniform cooling. Fin lengths of 25, 30 & 35 mm found most effective in reducing the temperature rise.
(Huang et al., 2006)	Conventional-PV panel	PV-PCM with fins inserted in 3 different configurations (i.e. EA, EB, EC)	Indoor experiment using Solar simulator for about 4 h	EA having max 2 fins, EB with max 32 fins, EC with strip Al matrix	RT25, 26.6 °C/232 kJ/kg & GR40, 43 °C/82 kJ/kg	–	More than 30 °C temperature drop achieved for each configuration.

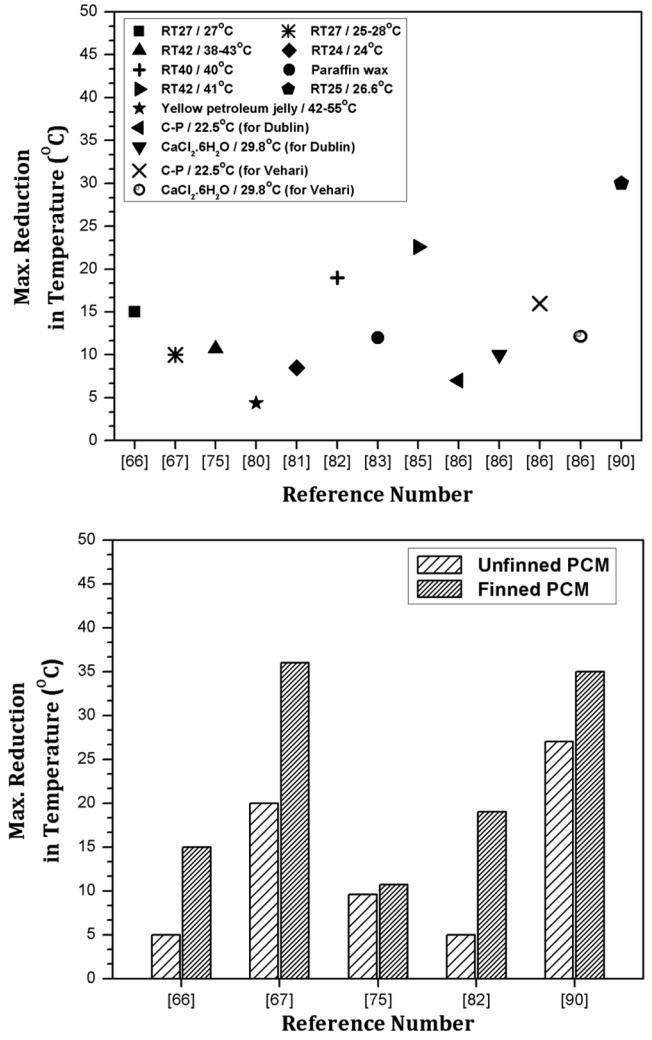


Fig. 11. Maximum reduction in PV-temperature using finned-PCM (top) and comparison with unfinned-PCM (bottom) achieved by different researchers.

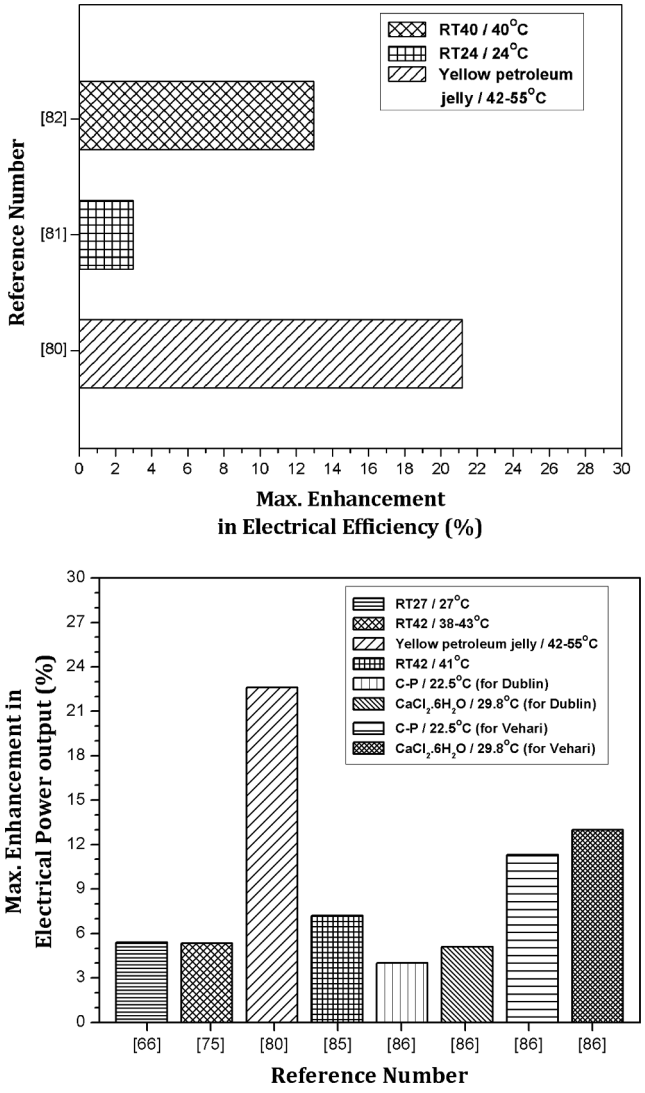


Fig. 12. Maximum enhancement in electrical efficiency (top) and electrical power (bottom) using finned-PCM as compared to conventional-PV achieved by different researchers.

which enhanced solar radiations up to 1.55 times. Composite-PCM brought more drop in PV-temperature than by pure-PCM. This value was 65 °C & 62 °C during indoor & outdoor experiment respectively.

Japs et al. (2016) investigated performance of pure & compound PCMs for PV cooling. Both PCMs had same T_m, but compound-PCM had 12 times higher thermal conductivity as compared to pure-PCM, resulting from graphite addition. Compound-PCM showed slower rate of rise in PV-temperature than by pure-PCM after melting. From technical & economical viewpoint, performance difference compared to ref-panel was insignificant as only compound-PCM panel had higher economic yield than ref-panel but only for 2 of 32 days. Atkin & Farid (Atkin and Farid, 2015) compared 4 different techniques (A, B, C, D) for PV thermal management. Case 'A' had no cooling mechanism. Case 'B' had PCM infused with graphite (thermal conductivity of pure-PCM enhanced by about 66%) at its back. Case 'C' had finned heat sink for passive cooling while case 'D' had finned heat sink as well as PCM infused with graphite. Case 'D' showed maximum overall efficiency among all other cases due to thermal effects enhanced by graphite & fins.

Luo et al. (2017) presented solution for low thermal conductivity of PCM encountered during PV cooling. Expandable graphite was added in RT28, resulting in very thermally conductive (7.571 W/mK) compound-PCM. PV-temperature dropped by 25 °C in PV/PCM panel, was maintained for long duration. Maximum enhancement achieved in power output was 11.5%. Hachem et al. (2017) prepared combined-PCM by combination of pure-PCM (white petroleum jelly), Cu & graphite powders in mass % of 70, 20 & 10, respectively. Experimental results revealed that efficiency enhancement by combined-PCM panel was higher than by pure-PCM (5.8 & 3% respectively). Overall, use of combined-PCM instead of pure-PCM, was found more effective due to its higher thermal conductivity. Sharma et al. (2017) compared performances of naturally cooled, PCM cooled, nano-PCM cooled BIPV with & without micro-fins respectively. CuO nanoparticles with 0.5 wt % were added in paraffin wax RT42 to form nano-PCM having higher thermal conductivity than pure-PCM. Effect of nano-PCM inclusion on temperature reduction was higher than by micro-fins & PCM. 12.5 °C average drop in temperature followed by power increase of 6.25%, was obtained through micro-finned nano-PCM BIPV system.

Table 3 shows summary of proposed work by different researchers for PV cooling using composite-PCM.

2.6. Discussion

Fig. 15 (top) shows different methods like nanoparticle addition, insertion of metal turnings, mixing with expandable graphite etc. used by researchers for thermal conductivity enhancement. Maximum increase (i.e. 16.35 W/mK) in thermal conductivity of PCM infused with

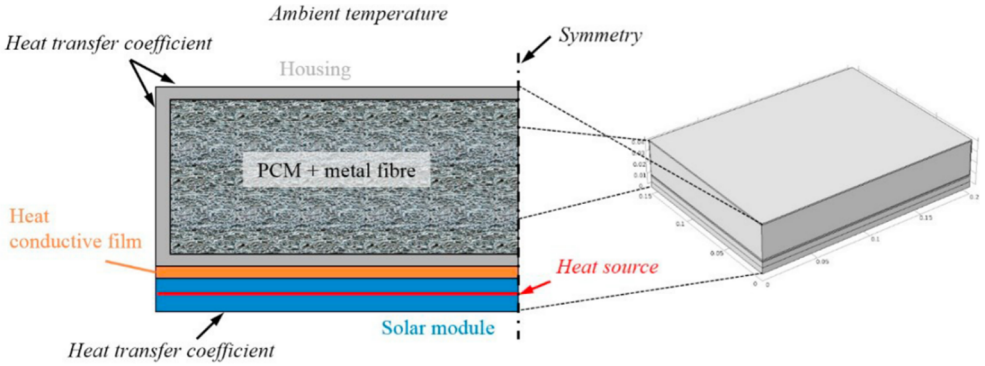


Fig. 13. 3D model for PCM module – concept and geometry. (Klemm et al., 2017)



Fig. 14. PCM embedded with metal turnings (Maiti et al., 2011).

graphite is achieved by (Atkin and Farid, 2015). From comparison of max. reduction in PV-temperature using pure & composite-PCM, it can be seen from Fig. 15 (bottom) that except (Japs et al., 2013), all researchers (Hachem et al., 2017); (Sharma et al., 2017); (Klemm et al., 2017); (Maiti et al., 2011) who used compound PCM found more temperature drop than by using pure-PCM due to its better temperature regulation. In case of (Japs et al., 2013), compound-PCM showed less temperature drop than pure-PCM as its heat storage capacity was compensated by 'k' enhancement but even after PCM melting, PV-temperature rise was at slower rate than pure-PCM. Overall, PCM with high thermal conductivity i.e. compound PCM is more effective in PV cooling in terms of temperature drop and slower temperature rise after PCM melting.

Fig. 16 shows maximum enhancement in PV power output using composite-PCM, achieved by different researchers. Max. enhancement (about 55%) is achieved by (Sharma et al., 2017) in which nano-PCM was tested with fins at panel rear for cooling. Reason is that although nanoparticles addition in PCM decreased heat absorbing capacity very slightly but it enhanced its thermal conductivity which increased heat conduction, thereby absorbing panel extra heat and so hindered its power decline trend.

2.7. PV with two PCMs separated by induced fins

Ho et al. (2017) evaluated performance of double layers of MEPCM for PV cooling using numerical simulation, and compared it with single MEPCM layer. In case of single MEPCM layer with T_m = 30 °C & 3 cm layer thickness, PV-cell performance was maximum in summer conditions but it reduced in winter conditions during which phase-change cycle could not be completed. Similarly, if lower melting point layer of MEPCM was used, PV-cell performance was good in winter but it was ineffective in summer as MEPCM melted earlier than required. Contrarily, double MEPCM layer showed good round-the-year performance, also reducing maintenance & other costs. Two MEPCM layers with T_m = 26 °C & 30 °C and combined layer thickness of 3 cm, were employed for PV cooling (Fig. 17). Order of MEPCM layers had very little impact on PV-cell performance. Optimum case was found when lower melting point layer had 0.5 cm thickness, while higher melting point layer had 2.5 cm thickness. This optimum configuration could give better thermal management in all conditions i.e. winter & summer. Low melting point layer played its role in winter condition while latent heat capacity of high melting point layer was used in summer conditions. Double layer configuration was found more effective than single MEPCM or without MEPCM layer configurations. It was also found more convenient and reduced maintenance as well as installation cost due to its round-the-year performance.

M.J Huang (Jun, 2011) reported a novel strategy for thermal regulation of photovoltaics through numerical simulation carried out using summer conditions. Different combinations each consisting of twoPCMs separated by metal fins, were used in simulation. 4 different PCMs (RT21, RT27, RT31, RT60) having melting points in range of 21 °C to 60 °C were investigated. Moreover, comparison was also made between two metal cells configurations (triangular & half circular cells) for separating 2 PCMs (Fig. 18). Total of 5 different combinations each consisting of 2 PCMs, were investigated. These combinations included RT27-RT21 (triangular cells), RT27-RT27 (triangular cells), RT31-RT27 (triangular cells), RT31-RT27 (half circular fins) and RT60-RT21 (triangular cells). First combination was found most effective as highest PV-temperature reduction was obtained by it. But its thermal regulation period was shorter than other combinations. In all cases, system was controlled under 30 °C leading to increase in PV-efficiency. Comparison of triangular & half circular cells showed insignificant difference in PV/PCM system performance although triangular cells reduced PV front temperature bit more than by half circular cells. Moreover, triangular cells also released PCM pressure inside container, better than half circular cells when PCM melted after expanding. Thermal regulation and temperature uniformity using 2 PCMs was found better than using single PCM.

2.8. Discussion

PV systems cooled with 2 PCMs at a time in comparison with single PCM, are more effective for round-the-year performance. Heat transfer regulation as well as PV-surface temperature uniformity is increased when 2 PCMs having different melting points are used. Also, unlike to single PCM where for different weather conditions PCM needs to be replaced, use of two PCMs has no such problems as low melting point

Table 3
Summary of proposed work by different researchers for PV cooling using composite PCM.

Ref no.	PV-panel specifications	Type of PV system	Nature of Work and Duration	PCM tested, melting point (^{0}C)/Latent Heat (kJ/kg)	Thermal conductivity enhancement $\Delta k = k_{comp} - k_{pure}$(W/mK)	Peak $T_{PV\text{-}ref}/T_{PV\text{-}PCM}$	Key Findings
(Japs et al., 2013)	Three 30 W monocrystalline panels	Compound PCM filled in bags attached to PV-panel rear	Outdoor experiment in summer at Paderborn, Germany from 15th July to 16th August 2013	Compound PCM, 153 kJ/kg	2.2	~54 °C/ ~49 °C	Pure & compound PCM reduced PV temp. by 10 °C & 7 °C respectively. After PCM melting, PV temp. raised at slower rate for compound PCM than pure-PCM.
(Klemm et al., 2017)	PV-module	PV-module with metallic-fiber structures in PCM	Numerical simulation for 3 days	RT25 HC with 90% porous metallic fiber structures, 26.6 °C/232 kJ/kg	–	78 °C/60 °C	Cell temp. reduced by 20 °C. Metallic fiber structures increased heat flow within PCM. 50 mm was most appropriate PCM thickness.
(Maiti et al., 2011)	10 W PV-panels with dimensions 0.46 × 0.29 (m)	PV-module with metal wax composite PCM	Indoor experiment using solar simulator & outdoor experiment with natural sunlight	Paraffin wax with packed metal turnings, 56–58 °C/255 kJ/kg	11.7	~90 °C/ ~73 °C ~80°C/~68°C	PV-surface temp. dropped by 17 °C & 20 °C for indoor & outdoor experiments.
(Atkin and Farid, 2015)	Two mono-crystalline panels, 0.22 × 0.235 (m)	Case A: ref-PV - Case B: PV with PCM infused graphite - Case C: PV with fin heat sink - Case D: PV with infused graphite & fin heat sink	Indoor experiment using insolation provided by two 500 W halogen lamps for 24 h	Paraffin wax RT40 (40 °C) infused with graphite	16.35	Case B: 77 °C/75 °C Case C: 77 °C/65 °C Case D: 77 °C/66 °C	PV-efficiency enhanced by 7.32%, 11.7% and 12.97% for cases B, C and D respectively, when compared to case A.
(Luo et al., 2017)	Two 100 W polycrystalline PV-panels, 1025 × 671 × 30 (mm)	PV with composite PCM cuboids attached at PV-panel back	Outdoor experiment in Guangzhou, China from 3rd to 7th March 2017 followed by numerical simulation	Paraffin (RT28) expandable graphite PCM, 27.17 °C/164.8 kJ/kg	~7.371	61.7 °C/ 57 °C	Cell temp. was kept below 50 °C for 3.33 h. Power output enhanced by 7.28%.
(Hachem et al., 2017)	Two 30 W polycrystalline PV-panels, 50 × 50 (cm)	Conventional-PV-panel & PV with combined PCM at back	Outdoor experiment at Al-Khyara, West Bekaa of Lebanon on 20th May 2015	Combined PCM (white petroleum jelly, Cu & graphite), 36–60 °C,k = 92.1 W/mK	95.20	62.2 °C/ 56.8 °C	Combined PCM due to its higher thermal conductivity, enhanced efficiency by 5.8% while it was 3% for pure-PCM.
(Sharma et al., 2017)	Conventional-PV-panel	Naturally cooled, PCM cooled, nano-PCM cooled BIPV with & w/o micro-fins	Indoor experiment using Solar simulator	RT42, 38–43 °C/174 kJ/kg, (0.5 wt% CuO added)	0.07	72 °C/62 °C	Effect of nano-PCM inclusion on temperature drop was higher than micro-fins/PCM alone. Micro-finned nano-PCM BIPV system reduced temp. by 12.5 °C and enhanced power by 6.25%.

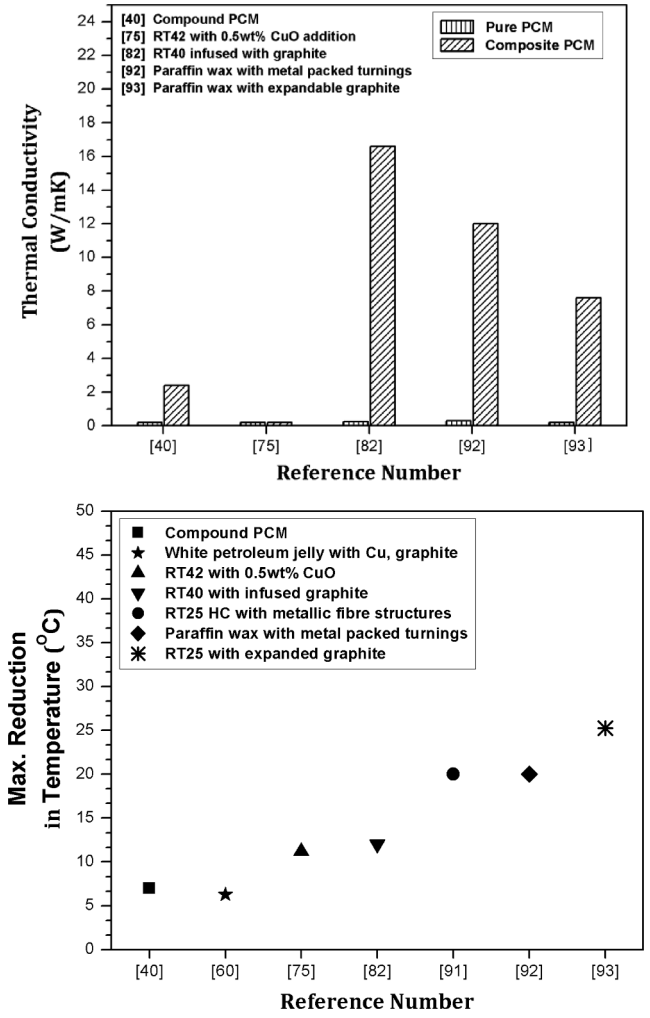


Fig. 15. Comparison of thermal conductivity (top) and resulting PV-temperature drop (bottom) by pure & composite-PCMs by different researchers.

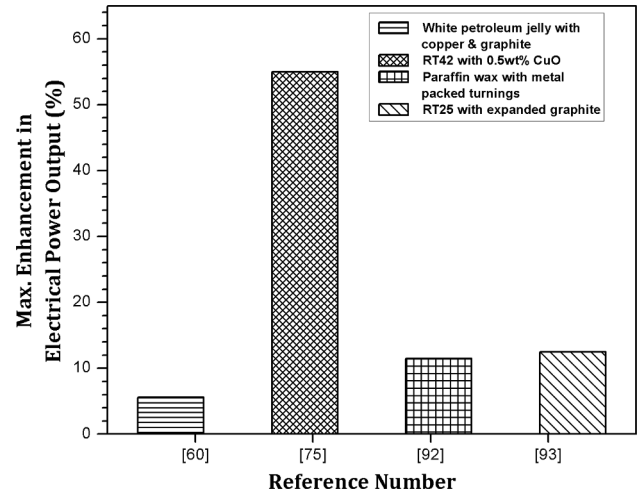


Fig. 16. Maximum enhancement in PV-electrical power output using composite-PCM as compared to conventional-PV achieved by different researchers.

PCM is more effective in winter while in summer, latent heat of PCM with high melting point plays its role. As a result, installation & maintenance costs are reduced to great extent. Hence, performance of PV systems with double PCM is better than those with single PCM in long run.

2.9. Hybrid PVT/PCM system

Hybrid PVT systems are capable of producing both electrical & thermal power output. Overall efficiency of such systems i.e. sum of electrical & thermal efficiencies, can be enhanced by integrating with PCM. Waste heat of PV-module is first absorbed by PCM in the form of latent heat as a result of which PV-surface temperature drops, leading to enhancement in electrical efficiency. After that, water flowing into tubes passing through PCM absorbs heat stored in it, resulting in enhanced thermal efficiency.

Electrical power output "E_{el}" is given as:

$$E_{el} = V_{OC} * I_{SC} * FF$$

Similarly, thermal power output "E_{th}" is given as:

$$E_{th} = m_f * c_{p,f} * (T_{f,o} - T_{f,i})$$

Now overall efficiency of PVT system is:

$$\eta_{pvt} = \frac{E_{el} + E_{th}}{A_c G}$$

2.9.1. Water-based hybrid PVT/PCM system

Radziemska & Kucharek (Klugmann-Radziemska and Wcisło-Kucharek, 2017) explained effects of varying PCM (Paraffin 42–44) layer thickness (2, 3, 4 cm) & water flowrate (32 & 80 dm^3/h) on performance of water-based PVT/PCM system. Best performance achieved with 2 cm PCM thickness & 32 dm^3/h water flow rate. But, PV/PCM without water cooling was found economically best option (due to technical difficulties while using water) as it reduced PV-temperature by 7 K while maintaining it for 5 h. Malvi et al. (2011) optimized various parameters like water flowrate, inlet temperature, PCM thickness to generate balance between PV & ST performance. By increasing PCM thickness, water flowrate and by decreasing water inlet temperature, PV performance enhanced while ST performance declined. At 2L/h water flow rate & 5 cm PCM thickness, PV power output enhanced by 9%. Optimum T_m of PCM depended upon climate. Fig. 19 shows construction of PV/ST/PCM system.

Browne et al. (2016) presented 3-day experimental data for 4 different configurations of PV-panels (Fig. 20), under outdoor conditions of Dublin. These configurations included ref-PV, PV with container, PVT & PVT/PCM systems. PVT/PCM showed best overall efficiency by increasing water temperature & better regulation of PV-panel temperature through uniform contact of PCM with panel.

Remarkable enhancements in PV-efficiency were found by Preet et al. (2017) after real-time experiment of PVT with water flowing at 3 different flow rates though Cu tubes inside finned-PCM. With increase in water flow rate, PV-temperature reduced further while PV-efficiency kept on increasing. About 53% temperature drop & 3 times PV-efficiency enhancement were found with PVT/PCM system.

Effects of water flow rate & inlet temperature on PVT/PCM performance were determined through numerical simulation results by Zhou et al. (2017). Different water flowrates (0.25, 0.5, 1, 5, 10 kg/s) & inlet temperature values (10, 15, 20, 25 °C) were investigated. Increasing water flowrate & decreasing its inlet temperature resulted in cell temperature reduction & PV-efficiency enhancement. Moreover, effect of water inlet temperature had more impact than its flowrate on PVT performance. Gaur et al. (2017) did numerical simulation of water-based PVT/PCM system with & without PCM using summer & winter conditions. PCM insertion in PVT system resulted in more cell temperature drop, electrical power & efficiency enhancement in summer than in winter. Moreover, effects of PCM mass & thickness as well as water flowrate on PVT/PCM performance showed optimal values found as 15 mm, 30 kg & 0.04 kg/s, respectively. Shukla et al. (2017) examined performance of water-based PVT system with composite-PCM (paraffin wax with fatty acid). PV-panel extra heat was absorbed by thermally conductive composite-PCM followed by water in tubes. PV-

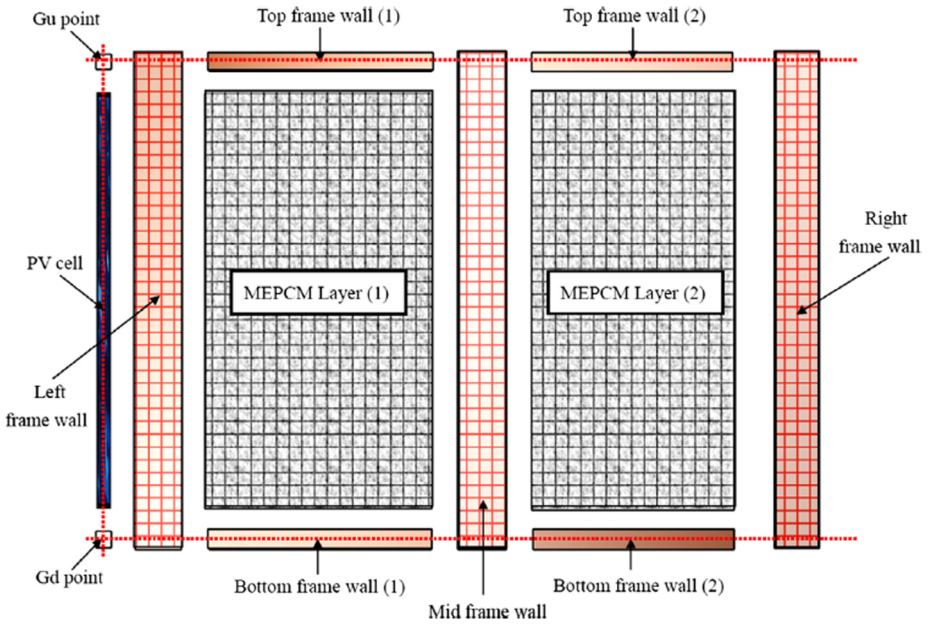


Fig. 17. Double layer PV/MEPCM cell – computational domains (Ho et al., 2017).

temperature dropped by 20 °C in hybrid PVT/PCM system as compared to ref-PV, leading to 21% increase in PV-efficiency. System overall efficiency also increased due to better heat regulation. Al Imam et al. (2016) investigated PVT/PCM performance with CPC in clear & semi-cloudy days for comparison. Overall efficiency of system was higher by about 10% in clear days than semi-cloudy days. PCM application resulted in gradually slower rise of panel temperature.

A novel PVT/PCM system was investigated experimentally by Browne et al. (2016). PCM used was capric: palmitic acid eutectic mixture which extracted extra heat from PV-panel, heating water flowing through pipes inside PCM. About 5 °C reduction in PV-temperature was determined. Hence, production of both electrical & thermal energy resulted in high overall efficiency of system. Su et al. (2017) evaluated that PVT thermal performance enhanced with higher T_m of PCM, but electrical performance declined. While lower T_m of PCM showed reverse performance. Optimal conditions were found as PCM of T_m = 40 °C with 3.4 cm thickness, by which cell temperature dropped by 20 °C leading to 15.2% increase in efficiency. Browne et al. (2015) compared performances of PVT/PCM system (mixture of capric & palmitic acid as PCM) with PVT through experimentation. Former absorbed 33% larger heat than latter as heat storage time also doubled. Hasan et al. (2016) investigated PVT/PCM system having nternally finned-PCM (RT42) through which water flowed at 0.027L/s in tubes, during winter conditions for 3-days. Average cell temperature dropped by 5 °C and PV-efficiency enhanced by 6% in PVT/PCM as compared to conventional-PVT. Thermal energy stored by water from PCM, was used for heating applications. Ho et al. (2014) attached MEPCM layer to rear of PV-panel made to flow on water surface. Varying parameters in numerical simulation were thickness (3 & 5 cm) and T_m (17 & 18 °C) of MEPCM. PV-efficiency remained constant as T_{amb} remained always less than 25 °C. Most effective case for releasing thermal energy to water during night, was by 5 cm MEPCM layer thickness having 17 °C T_m.

No Title (n.d.) compared through experimental investigation performances of three different PV systems i.e. ref-PV, PV with pipe

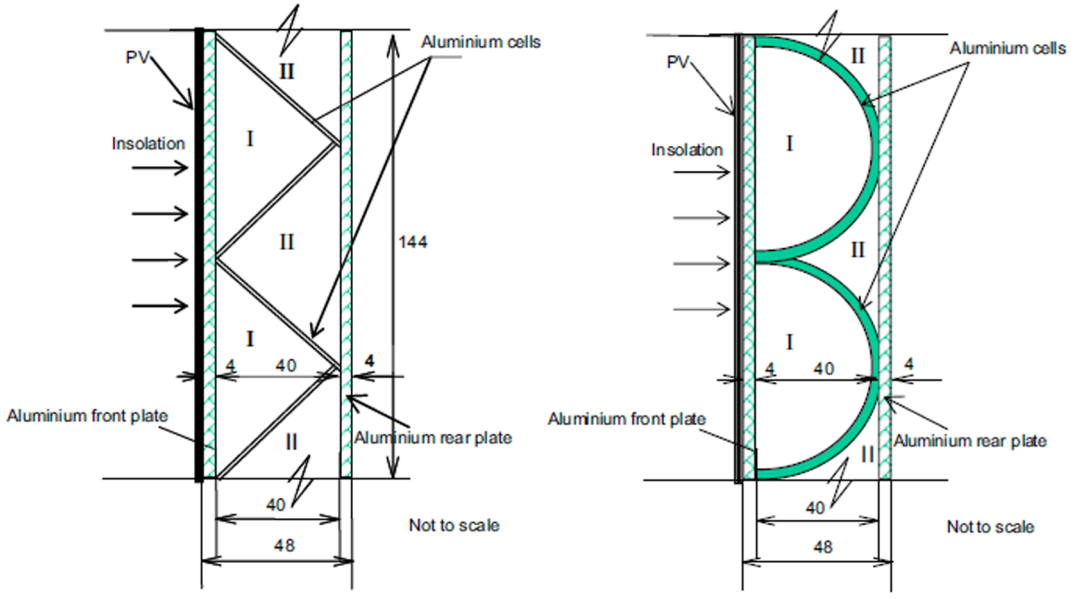


Fig. 18. Schematics of PV-system with 2 PCMs separated by triangular (left) & circular (right) cells (Jun Huang, 2011).

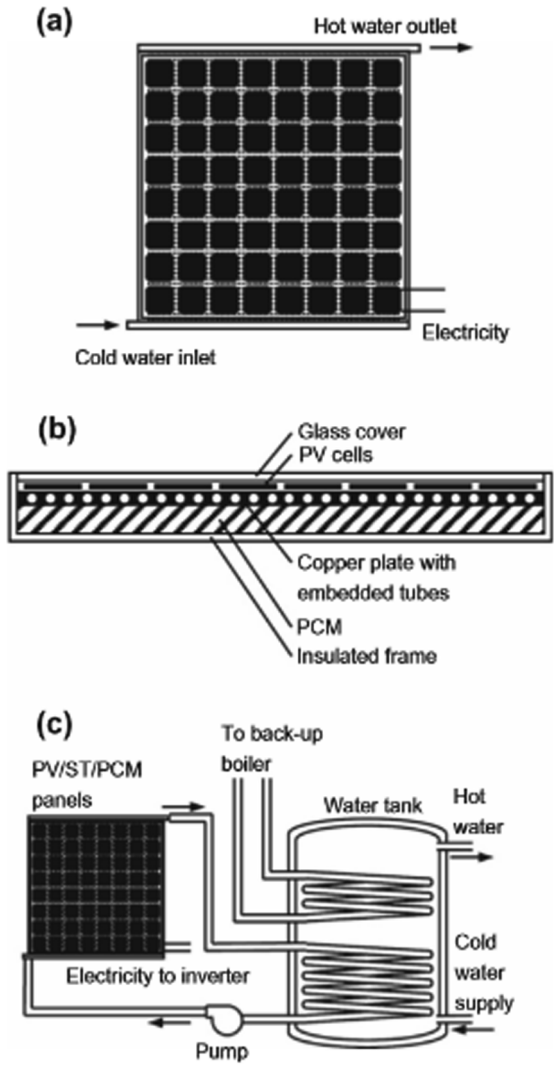


Fig. 19. Construction of PV/ST/PCM system from (a) plain view (b) cross-sectional view (c) illustration of active closed loop system (Malvi et al., 2011).

network for water & PV containing PCM as well as pipe network. Fatty acid eutectic was used as PCM which stored thermal energy. System 3 showed maximum PV-temperature drop during daytime when compared with system 2 and 1. This was due to extraction of PV extra heat by PCM as well as water. Yang et al. (2018) compared performances of PVT & PVT/PCM systems using solar simulator. Capric acid used as PCM was integrated along with Cu pipe network for water flow at PV-panel rear in PVT/PCM system. Significant drop in PV-temperature (15.8 °C) occurred between both systems due to absorption of panel extra heat by PCM followed by water in pipes. Ho et al. (2015) enhanced efficiency of untreated PV floating on water, by integrating MEPCM below PV-panel, in summer conditions. MEPCM of 2 different T_m (28 & 30 °C) and layer thicknesses (3 & 5 cm) were investigated. Best performance was achieved with 5 cm thickness of 30 °C MEPCM layer, for which PV-temperature reduced by 1.8 °C while generation output enhanced by 2.1%. Enhancement in generation output was higher at lower water flowrates.

Table 4 shows summary of proposed work by different researchers for PV cooling using water and PCM.

2.9.2. Discussion

When Fig. 21 is compared with Fig. 5, we see that combination of passive & active cooling techniques is more effective than passive cooling alone. Max. reduction in PV-temperature (i.e. about 50 °C) is achieved by (Zhou et al., 2017). Reason is absorption of PV extra heat by PCM (i.e. passive cooling) followed by water (i.e. active cooling) flowing at high flow rate through tubes inside PCM. In this way, more reduction in PV-temperature bringing it closer to 25 °C (ideal for PV) is possible.

From Fig. 22 (top), it can be observed that more than 15% increase in PV efficiency is achieved by all researchers except (Gaur et al., 2017); (Hasan et al., 2016) . Maximum enhancement found is about 32% by (Zhou et al., 2017) who used very high water flow rate along with PCM, resulting in more extraction of excess heat and so stopping decreasing trend of efficiency with temperature. Fig. 22 (bottom) shows that electrical power output enhancements up to 25% have been achieved with water-based hybrid PVT/PCM system. As efficiency of this system increases due to its high capacity of excess heat absorption, power output also increases accordingly making it very effective PV system.

In literature found on water-based hybrid PVT/PCM system, only a few researchers (Klugmann-Radziemska and Wcisło-Kucharek, 2017; Preet et al., 2017; Zhou et al., 2017) discussed effect of water flowrate on max. reduction in PV-temperature. According to (Klugmann-Radziemska and Wcisło-Kucharek, 2017; Zhou et al., 2017), more temperature drop is noticed with increasing flowrate (see Fig. 23), while (Preet et al., 2017) found the decreasing trend. It shows that flowrate of water is one of the important parameters in designing such PV systems.

2.9.3. Air-based hybrid PVT/PCM system

Fiorentini et al. (2015) proposed a novel approach of connecting HVAC system with solar PVT system integrated with PCM. PVT system generated PV-efficiency using required heat proportion while waste heat was stored in PCM for air-conditioning before air handling unit. This led to enhance system overall efficiency significantly. Bigaila and Athienitis (2017) modelled and simulated air-based BIPV system integrated with air-to-water heat pump which was further connected with a radiant PCM panel. PV-module generated electricity while waste heat was used to warm air which was supplied to heat pump followed by storing its heat in radiant PCM panel. This heat stored in PCM for short term was then used for space heating. Hence, a novel & effective hybrid system based on Canadian building was presented. Lin et al. (2016) analyzed space heating of house in winter by forcing warm air from beneath the PV-panels into house having walls integrated in separate phases with 3 PCMs (RT18HC, SP21E, SP24E). CTPE value was found in order of SP21E > RT18HC > SP24E. After optimization of air flowrate, type of PCM & its thickness, wall insulation etc., maximum CTPE found was 70%. Development of analytical model followed by experimental investigation of a novel HVAC system assisted with air-based PVT collector was done by Fiorentini et al. (2015). PV-panel was cooled by air underneath it, being used in HVAC system for direct air-conditioning or thermal storage in PCM. Mixture of salt hydrate having T_m = 22 °C was used as PCM for thermal storage. Presented design was for operation in both summer & winter conditions. Results of both simulation and experimentation had good agreement with each other.

Ren et al. (2017) used Taguchi method for evaluation of an air-based PVT system integrated with building. Hot air coming from beneath the PV-panel was moved inside building by fan installed. Building walls had PCM for thermal storage used for air-conditioning of building. Hence, PV-panel was cooled by air leading to increased electrical productivity while warm air stored heat into PCM inside walls. Among various control factors, air flowrate was found the most important one followed by type of PCM used. Aelenei et al. (2014) reported practical application of BIPV-PCM on a building in Lisbon during winter time. In prototype tested, air coming from ventilation system flowed between PV layer & PCM. Air flow and gap depth were varied to determine its effect on system behavior. Vents remained open during daytime. Heat from PV stored in air as well as PCM. This resulted in PV & overall efficiencies reaching 10 & 20%, respectively. Ceylan et al. (2016) discussed after experimentation the effect of using paraffin wax as PCM in

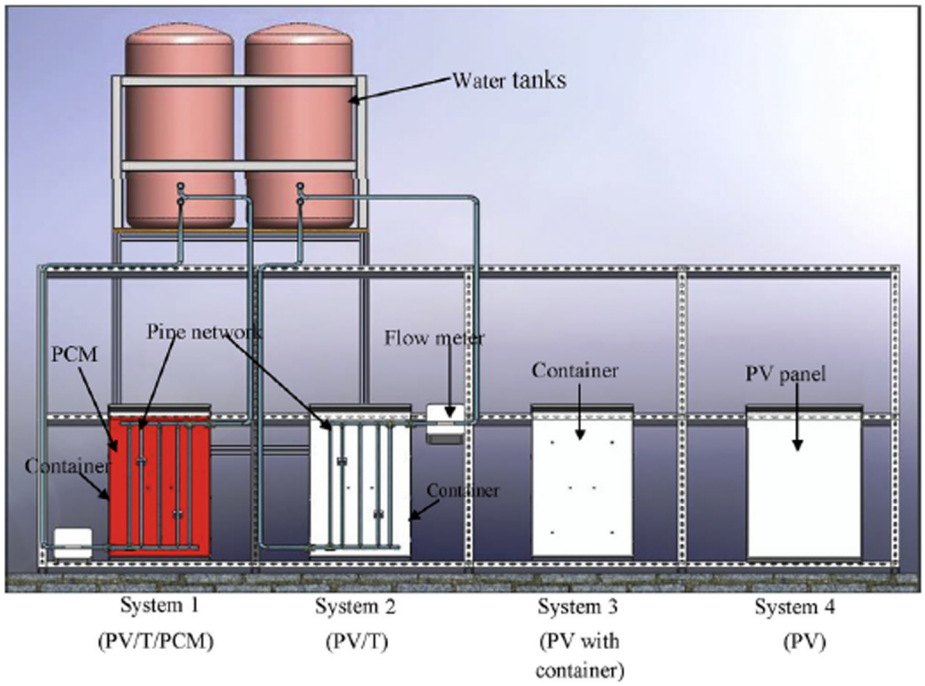


Fig. 20. Schematic diagram of experimental setup (Browne et al., 2016).

concentrated PVT system. Radiations coming from concentrator were focused on a sphere covered with broken PV-cells, and filled with PCM. Heat stored in PCM was used to dry the product in drying chamber. Panel efficiency increased from 5 to 11% as cell temperature reached to 37 °C with PCM.

Numerical study performed by Storage (2010) showed 7 °C drop in PV-module temperature in BIPV/T-PCM system as compared to same system without PCM. In both ventilated & non-ventilated systems studied, air temperature reduced due to presence of PCM in which solar heat was stored as latent heat. Although this resulted in lower thermal efficiency of BIPV/T-PCM system as compared to BIPV/T system but PV-efficiency increased due to reduced PV-temperature. W. Lin et al. (2014) evaluated through numerical study that ceiling ventilation system could be enhanced effectively, by integrating it with air-based PVT collector & PCM (Fig. 24). It showed better performance as compared to systems with PCM but without PVT & without PCM as well as PVT. Indoor thermal comfort increased using proposed system, thereby reducing the need of separate air-conditioning system.

Lin and Ma (2016) optimized different factors such as type of PCM and its thickness, flowrate of air, additional wall insulation etc. in air-based PVT collectors integrated with PCM. Optimization was done using Taguchi-Fibonacci method. PCM absorbed panel heat and passed it to the air. CPTE increased from 45.54 to 72.22% greatly showing the need of optimization. Su et al. (2017) examined effect of integrating PCM with air-based PVT solar collector through numerical simulation (Fig. 25). Position of PCM in PVT collector mattered as upper position showed overall efficiency enhancement of about 63% as compared to lower PCM position. Maximum PV-temperature drop of 7.4 °C resulted with PVT/PCM system withn3cm PCM thickness as compared to PVT system. This resulted in 10.7% higher overall efficiency than no PCM mode.

Xiang and Gan (2015) examined numerically and experimentally effect of fins inside PCM, thickness of PCM layer, air channel depth on controlling temperature rise of PV-panel. Heat removal rate from PCM was increased with lower PCM thickness & higher channel depth. But larger PCM thickness had larger heat storage capacity. Also, presence of fins provided improved control of temperature rise. Hence, air-based PVT/PCM system reduced panel temperature and conditioned indoor air for thermal comfort. Sudhakar et al. (2017) presented a model for efficiency improvement of PV. In this model, cool energy was stored in PCM during early morning when ambient temperature was low. Air cooled after passing through PCM tank, was flowed at low velocity to absorb the extra heat dissipated by PV-panel leading to reduction in its temperature during daytime. This resulted in increased electrical efficiency of PV-panels.

2.9.4. Discussion

In air-based PVT/PCM systems, both air & PCM absorb excess heat of PV-panel increasing its efficiency. Air after absorbing heat, can be further used for room ventilation. In this way, hybrid PVT/PCM system is formed which has the ability to produce both electrical as well as thermal energies.

2.9.5. Other hybrid PVT/PCM systems

Veeraragavan et al. (2014) integrated PCM (silicon) present in tapered tank, with PV for combined thermal storage & electricity production. Conc. sunlight entered through absorber where PCM absorbed major chunk of solar energy, and then left through tank emitter surface where PV-cells used leftover energy. So, energy storage & electricity production was obtained at low maintenance cost. Cui et al. (2016) determined that integration of PV-TE system with PCM resulted in increased performance as PCM reduced the temperature fluctuations in both PV-cell & TE modules. Qiu et al. (2015) did theoretical investigation of MEPCM slurry-based PVT module by developing a mathematical model. The proposed model showed very good prediction of energy performance. It was predicted that turbulent flow of MPCM slurry with Re = 3350 & 5 wt% conc. resulted in peak efficiency of proposed system. Experimental investigation of PV/T system based on MEPCM slurry was done by Qiu et al. (2016). Heat & power output increased with application of MEPCM slurry. At high Re = 3000 & 10 wt% conc. of MEPCM, net efficiency of PV/T module was optimized (i.e. in range of 80.8–83.9%). Experimental investigation of PV-PCM-TE system was also done by T. Cui et al. (2017) after designing such a novel system (Fig. 26). Requirement of maintaining system working temperature at desired level was fulfilled by PCM integration. This system showed higher efficiency due to better utilization of solar energy when

Table 4
Summary of proposed work by different researchers for PV cooling using water and PCM.

Ref no.	PV-panel specifications	Type of PVT system	Nature of Work and Duration	Water flowrate	PCM tested, melting point (^{0}C)/ Latent Heat (kJ/kg)/Layer thickness	Peak $T_{PV\text{-}ref}/T_{PVT\text{-}PCM}$	Key Findings
(Klugmann-Radziemska and Wcisło-Kucharek, 2017)	10 W PV-module, with dimensions 340 × 280 × 17 (mm)	PVT/PCM with water flowing in tubes inside PCM	Indoor experiment using Solar simulator for almost a day in Poland	32 dm^3/h 80 dm^3/h	Paraffin 42–44 (MERCK), 42–72 °C/ 130 kJ/kg/2, 3, 4 cm	–	With 2 cm PCM thickness and 32 dm^3/h water flowrate, PV-module performed at its best, increasing electrical power by 6.93%.
(Malvi et al., 2011)	PV-panel with surface area 1 × 1 (m^2)	PV with water flowing in copper tubes in PCM layer	Numerical Simulation for 10 days	2 L/h	Paraffin wax, 28 °C/210 kJ/kg/5 cm	–	At water flowrate of 2 L/h and PCM thickness of 5 cm, power output of PV enhanced by about 9% when compared to ref-PV.
(Preet et al., 2017)	100 W silicon-based PV-panel	PVT with water flowing in copper tubes through finned-PCM	Outdoor experiment at Gurdaspur, India for one day	0.013, 0.023, 0.031 kg/s	Paraffin wax RT30, 28 °C/222 kJ/ kg/7 cm	85 °C/63 °C	Cell temperature dropped by 24 °C (i.e. 54%) at 0.023 kg/s.
(Zhou et al., 2017)	PV-module	PVT/PCM with water	Numerical Simulation using conditions of Changsha, China	0.25, 0.5, 1, 5, 10 kg/s	PCM, 45 °C	70 °C/~33 °C	PV-cell temp. dropped by increasing water flowrate and decreasing its inlet temp. PV temp. reduced by max 47.2 °C resulting in 32% power output increase.
(Gaur et al., 2017)	110 W monocrystalline semi-transparent PV-panel	PVT collector (water) incorporated with PCM	Numerical simulation using conditions of 8th July & 20th Feb in Lyon, France	0.04 kg/s	OM37, 37 °C/211 kJ/kg/5 cm	–	Performance was better in July (summer). PCM thickness and mass as well as water flowrate had optimal values of 0.015 m, 30 kg and 0.04 kg/s respectively.
(Shukla et al., n.d.)	160 W poly-crystalline PV-panel, 1270 × 711 (mm)	Water-based PVT with composite PCM	Outdoor experiment in March 2017 at Bhopal, India	–	Paraffin wax with mixture of fatty acid, 47–58 °C	65 °C/47 °C	PV-panel temp. dropped by 20 °C leading to 21% increase in electrical efficiency. System overall efficiency also increased by 50%.
(Su et al., 2017)	PV with dimensions 1.8 × 1 (m)	PVT solar collector with PCM	Numerical simulation using conditions of August 2014 in Nanjing, China	0.03 kg/s	PCM of different melting points, 210 kJ/kg/5 cm	100 °C (PVT without PCM)/ 80 °C (PVT with PCM)	Cell temp. dropped by 20 °C leading to enhancement in electrical power by 14.4% and efficiency by 15.2%. Optimal PCM m.p was 40 °C with 3.4 cm thickness.
(Hasan et al., 2016)	40 W polycrystalline panels, 53 cm × 63 cm	PV-PCM with fins inserted and water flowing in tubes in PCM	Outdoor experiment at Al-Ain, UAE from 30/1/15 to 2/2/15	0.027L/s	RT42, 38–43 °C/145 ± 7.5% kJ/kg	53 °C/44 °C	Average drop in PV temp. was 5 °C. Electrical efficiency enhanced by average 6%.
(Yang et al., 2018)	100 W PV-module with dimension 0.648 m^2	PV with PCM and absorber plate attached to piping for water	Indoor experiment using Solar simulator	0.15 m^3/h	Capric acid, 30.1 °C/158 kJ/kg	–	Significant drop in PV-temperature (i.e. 15.8 °C) achieved.

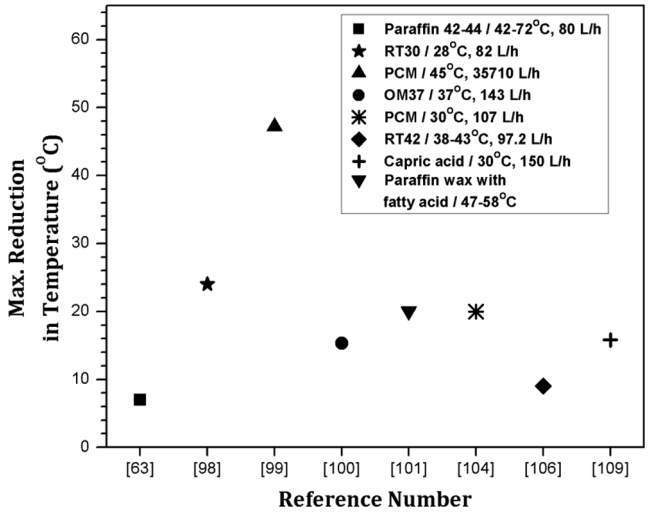


Fig. 21. Maximum reduction in PV-temperature using water at specific flowrate and PCM as compared to conventional-PV achieved by different researchers.

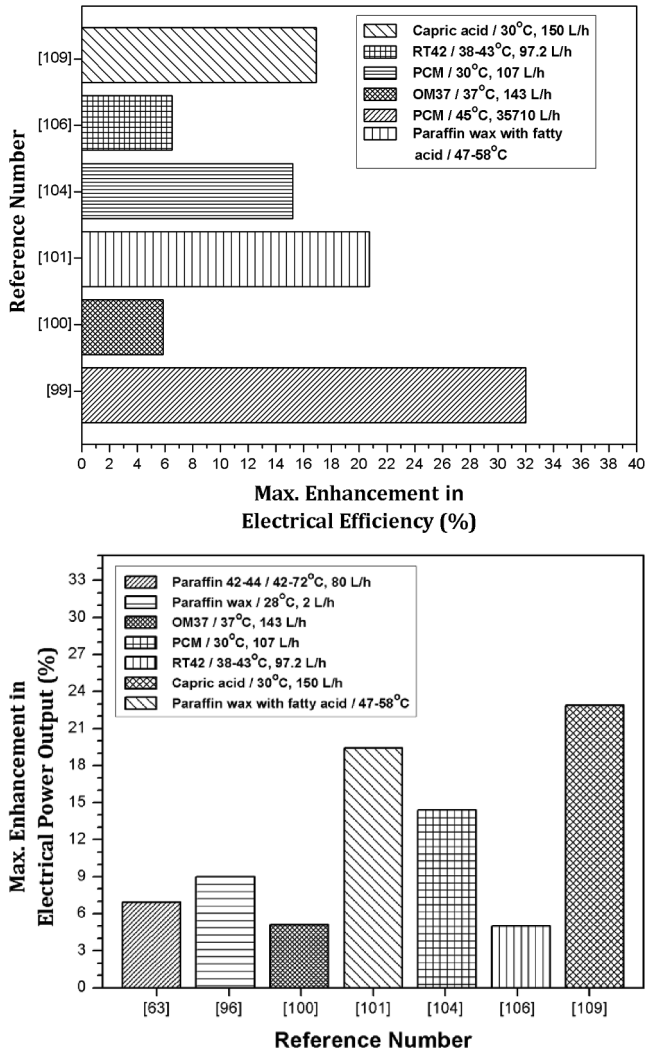


Fig. 22. Maximum enhancement in electrical efficiency (top) & power (bottom) using water at specific flowrate and PCM as compared to conventional-PV achieved by different researchers.

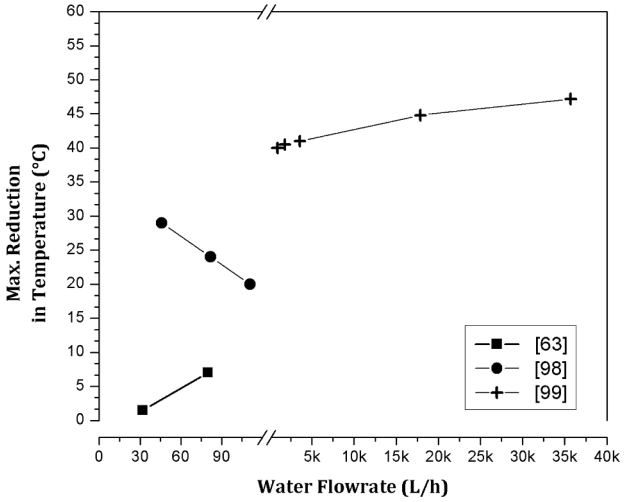


Fig. 23. Water flowrate vs Maximum reduction in PV-temperature using water & PCM as compared to conventional-PV achieved by different researchers.

compared to pure PV system.

Veeraragavan et al. (2017) developed an analytical model to study lateral heat losses in tank of SISTPV system. It was found that increasing SR to a certain value increased thermal efficiency. Similarly, increasing AR, length of storage tank, TR resulted in increase in thermal efficiency of SISTPV system.

2.9.6. Discussion

PCM can be used for PV system when it is integrated with other systems like thermo-electric module etc. which results in excess heat absorption in its latent heat while aiding other system integrated with photovoltaics.

2.10. Selecting suitable PCM for PV-module

The purpose of using PCM for PV cooling is that it can absorb waste heat in the form of its latent heat and can maintain cell-temperature constant for some duration. Hence, PCM having high value of latent heat and thermal conductivity should be selected. As normally, PCMs have low thermal conductivity, so in this regard, thermal conductivity enhancers can be used. Melting point of PCM needs a lot of importance while selecting PCM. Although there is not a particular criterion for its selection but if low melting PCM is used, it can maintain PV-surface temperature near to ideal PV-temperature of 25 °C but for short duration. If melting point is high, then although PV-surface temperature is not close to 25 °C but it is kept constant at some higher value for longer duration. Moreover, while selecting PCM, geographical location and its climatic conditions are also the parameters to be taken care of. Other than that, thickness of PCM layer can be manipulated if still there is some difficulty in selecting PCM with right melting point. For better heat regulation & round-the-year performance, 2 PCMs with different melting points can also be used at the same time.

3. Hybrid PVT system with nanofluids

Nanofluids have very high heat absorbing characteristics due to their enhanced thermal conductivity as compared to their base fluids. Hence, they can be used as efficient coolants for photovoltaics, absorbing large amount of waste heat due to which PV-surface temperature reduces significantly, getting closer to ideal operating temperature (i.e. 25 °C). This leads to enhanced electrical power output. Moreover,

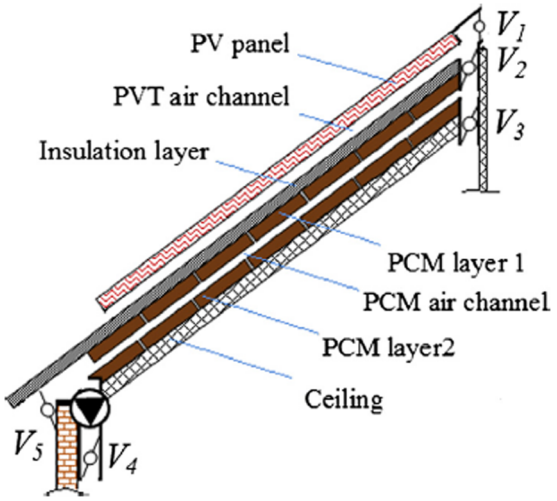


Fig. 24. PVT collector in combination with PCM-enhanced ventilation system (Lin et al., 2014).

heat absorbed by nanofluid can be further utilized to make hybrid PVT system having higher overall efficiency than by base fluid. It is also found that nanofluid can be considered as spectral filter for photovoltaics, as they can selectively absorb infrared radiations falling over PV-surface. In following sub-sections, both heat absorbing & spectral filtration characteristics of nanofluids in photovoltaics, are explained by work of different researchers.

3.1. Hybrid PVT system with nanofluid as thermal absorber/coolant

Al-Waeli et al. (2017) performed real outdoor experiment using SiC-water for PV cooling purpose. SiC addition in water increased its thermal conductivity by 8.2% which increased its heat transfer characteristics. Nanofluid-based PV/T system was superior to both water-cooled and standalone PV system as PV-temperature reduced by 16 °C while electrical power & efficiency enhancements were 57% & 24.12% respectively. Sardarabadi and Passandideh-Fard (2016) compared 3 different metallic oxides (TiO_2; ZnO, Al_2O_3)/ water nanofluids for efficiency improvement of PV/T system in real outdoor conditions. Peak PV-efficiency enhancements for these nanofluids were 6.54, 6.46 & 6.36% respectively. Best thermal efficiency of PV/T system was achieved with ZnO-water nanofluid. Fig. 27 shows different types of heat transfer mechanisms involved in PVT system.

Hussien et al. (n.d.) designed PV/T system in solar laboratory and tested addition of Al_2O_3 nanoparticles (0.3%wt) in water for its cooling purpose. Flowrate of nanofluid was kept constant at 0.2L/s. Maximum temperature drop of 36.9 °C on PV-surface & 51.2% efficiency improvement showed excellent heat transfer characteristics of Al_2O_3 nanofluid. Ghadiri et al. (2015) did experimental investigation using ferrofluid (Fe_3O_4-water) with 2 different concentrations (1 & 3 wt%) under 2 different solar radiations (600 & 1100 W/m^2). At higher radiation & 50 Hz magnetic field, 3 wt% concentration increased PV & thermal efficiency of PV/T system more than by 1 wt%. Maximum PV-efficiency enhancement was 37% with 3 wt% conc. Sardarabadi et al. (2014) connected sheet & tube collector with PV system for its cooling by SiO_2-water nanofluid at 2 different conc. (1 & 3 wt%). PV-panel temperature raised to its peak value of 68 °C without cooling, but nanoparticles addition reduced PV-temperature to peak 50 °C, improved PV-efficiency from 10 to 11% by increasing conc. from 1 to 3 wt%. Overall efficiency enhanced both energetically & exergetically. Hamdan and Kardasi (2017) cooled PV-module from backside by 2 types of nanofluids (Al_2O_3 & CuO/water) under real conditions in Jordan. CuO due to its higher thermal conductivity, performed better than Al_2O_3 as PV-efficiency enhanced by 35 & 25% respectively. Optimal concentrations for best PV performance were 0.4% Al_2O_3 & 0.6% CuO.

Karami and Rahimi (2014) compared 2 channels (straight & helical) for Boehmite nanofluid ($AlOOH{\cdot}xH_2O$) on module backside. Conc. used were 0.01, 0.1 & 0.5 wt% with flowrates varying from 0 to 400 mL/min. Helical channel geometry performed better than straight one due to more metal contact with nanofluid. 0.1 wt% showed best performance followed by 0.01 & 0.5 wt%. After 200 mL/min flowrate, performance declined as flow transited from laminar to turbulent. Maximum increase i.e. 37.67% in electrical power was found using helical channel & 0.1 wt %. Ragab (Performance of cooling photovoltaic cells using nanofluids performance of cooling photovoltaic cells using nanofluids, 2016) did numerical simulation of PV, cooled by Boehmite nanofluid (0.01, 0.1 & 0.5 wt%) flowing through straight channels at different flowrates. Maximum temperature reduction of 24 °C & electrical power increase of 21.87% was obtained with 0.1 wt% concentration. These were achieved at flowrate less than 200 mL/min after which flow transited from laminar to turbulent. Lari and Sahin (2017) performed CFD analysis to compare PV/T system based on nanofluid (Ag-water containing surfactant) with water. Nanofluid lowered cell temperature 2 °C more than that by water while electrical power increased by 0.9% respectively. Reason behind this was presence of Ag nanoparticles which due to their high thermal conductivity, increased heat transfer ability of water. Nanofluid-based PV/T system was economical in long-term due to its high efficiency.

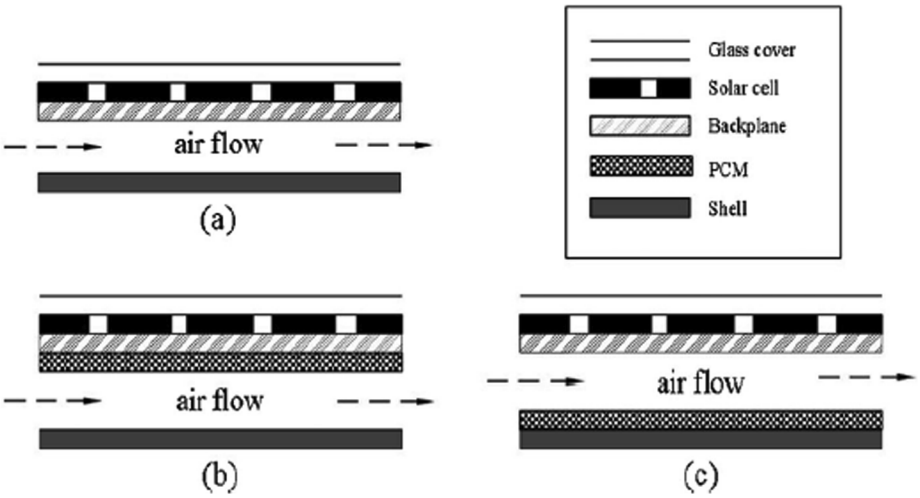


Fig. 25. Different positions of PVT collector with PCM – a sectional view (Su et al., 2017).

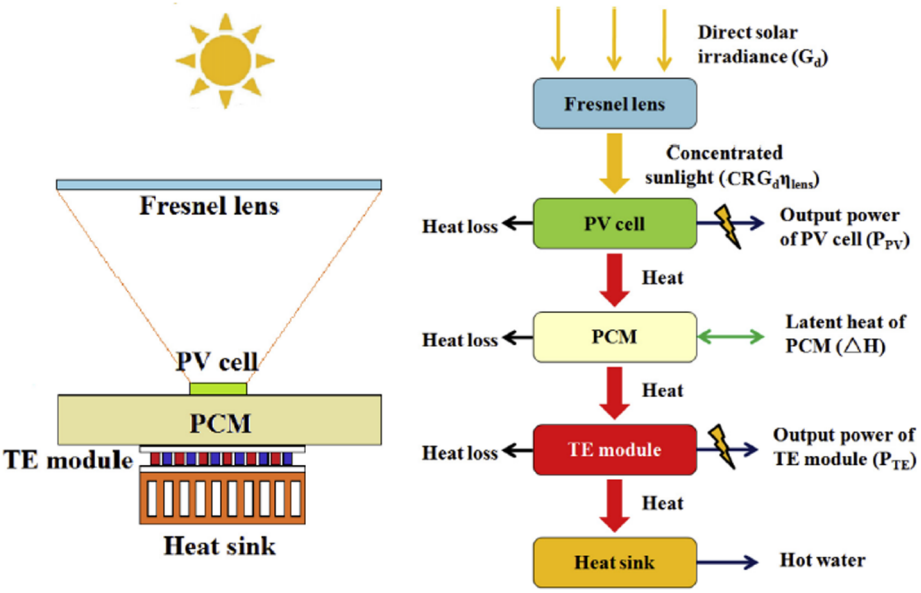


Fig. 26. Schematics of PV-PCM-TE hybrid system (Cui et al., 2017).

Al-Waeli et al. (2017) compared performances of uncooled PV, water and nanofluid-cooled PV/T systems using solar simulator. Nanofluid performed best for which 3 nanoparticles (Al_2O_3, CuO, SiC) with 5 vol conc. (0.5, 1, 2, 3, 4%) were tested. 0.5 vol% showed maximum stability while 4 vol% resulted in maximum thermal conductivity. Among other nanoparticles, SiC resulted in highest electrical power, PV & thermal efficiency due to its highest thermal conductivity. Radwan et al. (2016) determined effects of nanoparticle type (SiC & Al_2O_3), its volume fraction, Re & conc. ratio on performance of nanofluid-cooled PV/T system. Increasing volume fraction up to 4%, decreasing Re and increasing C.R resulted in higher temperature drop & electrical power of PV. SiC due to its better thermal conductivity was found more effective than Al_2O_3. Soltani et al. (2017) compared performances of silica & Fe_3O_4 nanofluid in hybrid PV/TE system with naturally cooled-PV panel. Silica-nanofluid due to its higher thermal conductivity, performed better than Fe_3O_4 nanofluid. Maximum electrical power improvements were 54.29 & 52.40% for silica & Fe_3O_4 nanofluids respectively, while PV-efficiency enhanced by 3.35 & 3.13% respectively. Hassani et al. (2016) stated through numerical simulation results that with nanofluid (CNT-water), temperature of ref-PV reduced from 147.7 °C to 59.4 °C. Remarkable performance achieved as electrical power & PV-efficiency enhanced by 53.84 & 61.5%, respectively. Ebaid et al. (2018) cooled PV with 2 nanofluids (Al_2O_3 in water-

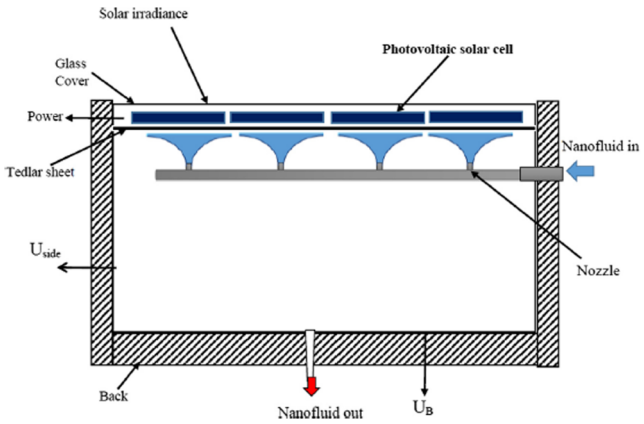


Fig. 28. PVT collector with impingement of jet nanofluids (Hasan et al., 2017).

polyethylene glycol mixture & TiO_2 in water-cetyltrimethylammonium) at 0.01, 0.05 & 0.1 wt% nanoparticles conc. for different flowrates. Peak PV-temperature reduced from 57.5 °C to 39.5 °C & 40.5 °C by Al_2O_3 & TiO_2-based nanofluids respectively, at 0.01% conc. & 3000 mL/min flowrate. Performance increased by increasing conc. & flowrate

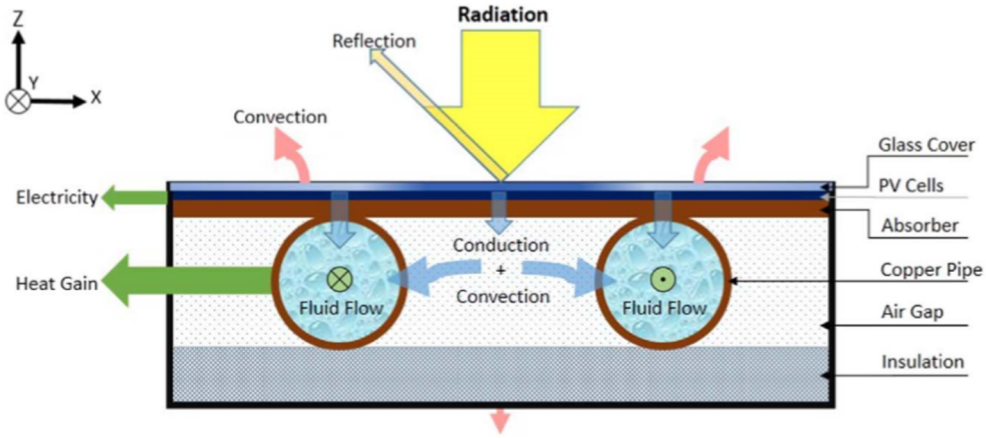


Fig. 27. Different types of heat transfer mechanisms in PVT system (Sardarabadi and Passandideh-Fard, 2016).

while Al_2O_3 was more effective than TiO_2 for all conc. & flowrates considered. Hasan et al. (2017) investigated influence of jet impingement of 3 nanofluids (SiC, TiO_2, SiO_2/water, 1 wt%) on power output & efficiency of PV/T system at flowrates varying from 0.05 to 0.167 kg/s (Fig. 28). SiC showed best performance followed by TiO_2 & SiO_2 and it enhanced with increasing mass flowrate. PV-efficiency enhanced by 60%, 55% & 47% for SiC, TiO_2 & SiO_2-based nanofluids respectively.

Elmir et al. (2012) used Al_2O_3-water nanofluid for cooling of PV/T system. Results of numerical simulation showed that nanofluid performed better than water due to its enhanced thermal conductivity. Heat transfer rate increased by increasing nanoparticles conc. and by decreasing Re. Xu and Kleinstreuer (2014) did CFD analysis of CPV/T system cooled by Al_2O_3-water nanofluid in channels on panel back. Cell efficiency increased by increasing Re up to 20,000 and then insignificantly increased at 4% vol. fraction. Inlet fluid temperature had reverse effect on performance of CPV/T system. Best channel height was found to be 10 mm. Xu and Kleinstreuer (2014) stated through numerical simulation results that triple junction cells were more effective than Si cells for CPV/T system. With increasing inlet temperature and decreasing outlet temperature of nanofluid, electrical efficiency decreased while thermal & overall efficiency increased. Fig. 29 shows schematic diagram of typical CPV/T system (see Fig. 30).

Khanjari et al. (2017) reported that by increasing solar irradiation from 200 to 800 W/m^2, PV-efficiency of nanofluid-based PV/T system decreased from 11.4% to 10.3%, while by increasing inlet temperature from 293 K to 323 K, it declined from 11.2% to 9.6%. Another study by Khanjari et al. (2016) revealed that by increasing vol. fraction of nanoparticles from 1 to 10%, electrical efficiency of PV/T system enhanced by 0.28 to 1.83% for alumina-water, & by 0.69 to 3.9% for Ag-water nanofluids, respectively when compared to pure water. Similar kind of trend was found for thermal efficiency. Effect of mass fractions (0 to 10% of nanoparticles) of various nanoparticles (ZnO, Al_2O_3, TiO_2, SiO_2) on entropy generation of nanofluid-based PV/T system was experimentally & numerically examined by Paper (2017). Entropy generation decreased with increasing mass fraction of nanoparticles in nanofluids. Metallic nanofluids were found effective as performance of different nanoparticles was in order of Al_2O_3 > TiO_2 > ZnO > SiO_2. Sdqho et al. (n.d.) experimentally investigated performance of PV/T system with Al_2O_3-water nanofluid by varying conc. (1 & 3 wt%). Thermal efficiency & system power was found better for 3 wt% conc. as compared to 1 wt% conc. Increase in mean temperature of nanofluid decreased current output of system. Moreover, concentration of nanoparticles had insignificant effect on overall efficiency of PV/T system.

Sathieshkumar (2017) cooled PV by flowing hybrid nanofluid (0.2 wt% $CuTiO_2$-water) at varying flowrates (0.01, 0.02 & 0.03 kg/s) in baffle arrangement of collector. Max. enhancements i.e. 5.8% & 3.6% in PV & thermal efficiencies respectively, as compared to water-based system were found at 0.02 kg/s. Michael and Iniyan (2015) replaced tedlar layer of PV-module with Cu sheet to reduce thermal resistance for maximum solar heat absorption by nanofluid (CuO-water) in thermal collector. This decreased power output of PV-module. Glazing trapped some radiations resulting in higher cell temperature. PV-efficiency of hybrid PV/T collector decreased by 31.18% & 15.14% with & without glazing, respectively. Overall efficiency was higher than ref-PV due to higher thermal efficiency obtained. Gangadevi et al. (2013) examined Al_2O_3-water (50 nm, 0.5 vol%) nanofluid in flat plate collector attached to PV-panel. Nanofluid due to its enhanced heat transfer characteristics increased PV-efficiency & collector's thermal efficiency.

Sardarabadi et al. (2017) experimentally investigated performances of ref-PV & PVT system with ZnO/water (0.2 wt% conc.). Cell temperature reduced by maximum 7 °C leading to 8% enhancement in PV-efficiency. Chandrasekar et al. (2013) employed a novel strategy for PV cooling by placing cotton wick structures at panel back, and dipped free ends in nanofluid. Panel cooled due to capillary action in cotton by nanofluid. Temperature dropped by 6 °C & 11 °C as compared to ref-PV while PV-efficiency improved from 9 to 9.5% & 9.7% with CuO & Al_2O_3/water nanofluids respectively. Hussien and Noman (2015) performed indoor investigation of PV cooling by nanofluid (Al_2O_3-water) which was circulated through pipes at panel back at volume conc. of Al_2O_3 varied as 0.1, 0.2, 0.3, 0.4 & 0.5%. Thermal efficiency increased by increasing conc. up to 0.5% but PV-efficiency enhanced by maximum 51.25% at 0.3% conc. due to drop in cell temperature by 36.9 °C as compared to system with water. Al-Shamani et al. (2016) tested PV/T collector with 3 different nanofluids (SiC, TiO_2, SiO_2 -water) having 1 wt% conc., at flowrates varying from 0.068 to 0.170 kg/s. With increase in mass flowrate, both PV & thermal efficiencies as well as temperature reduction also increased, showing best performance at 0.170 kg/s. PV efficiencies achieved were in order of 13.52%(SiC) > 10.98%(TiO_2) > 10.3%(SiO_2).

Karami and Rahimi (2014) investigated effects of flowrate (0 to 300 mL/min) & nanoparticles conc. (0.01, 0.1, 0.3 wt%) on performance of Boehmite nanofluid in rectangular micro-channels attached at PV rear. Performance enhanced with increasing flowrate & decreasing nanoparticles conc. At higher conc., problems like nanoparticles agglomeration occurred. With 0.01 wt% & 300 mL/min flowrate, cell temperature reduced by 29.8 °C & electrical power enhanced by 27%.

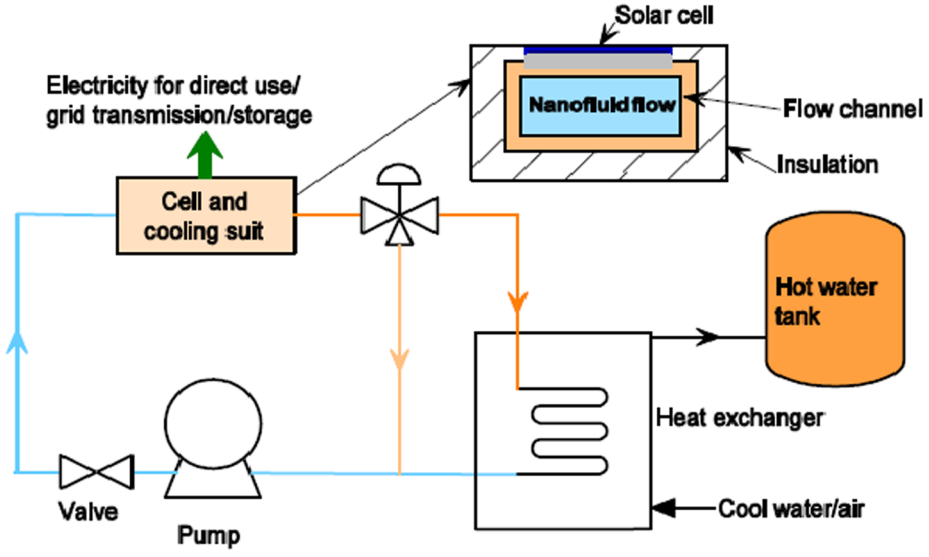


Fig. 29. Schematic diagram of typical CPV/T system (Xu and Kleinstreuer, 2014).

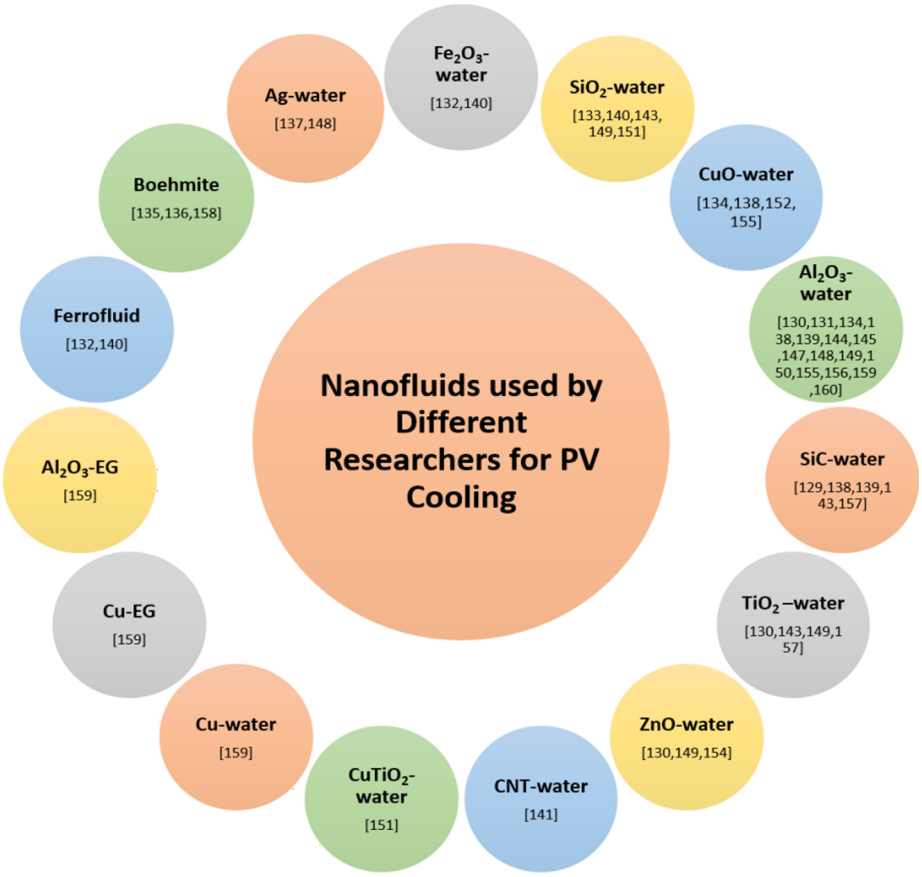


Fig. 30. Nanofluids used for PV cooling by different researchers in the available literature.

Rejeb et al. (2016) compared performances of PVT system based upon nanofluid & its base fluid. 4 different nanofluids were tested (Cu-water, Al_2O_3-water, Cu-EG, Al_2O_3-EG) with conc. varied as 0.1, 0.2, 0.4 wt%. Water-based nanofluids showed more enhancement in PV-efficiency than EG-based nanofluids. Cu-nanoparticle enhanced efficiency more than Al_2O_3. Also, performance increased with increasing conc. from 0.1 to 0.4 wt%. Lelea et al. (2015) reported that nanofluids formed with highly thermally conductive nanoparticles like Al_2O_3, could be a good cooling solution for CPV-cells. Maximum lowering of temperature obtained with low Re of nanofluids.

Table 5 shows summary of proposed work by different researchers for PV cooling using nanofluid as coolant.

3.2. Discussion

When Fig. 31 (a & b) is compared with Fig. 21, it can be observed that if active cooling is done with nanofluid instead of base fluid (water), more prominent results of temperature reduction are achieved. With nanofluid only, maximum reduction in PV-temperature about 60 ˚C is achieved by (Hassani et al., 2016) by the use of CNT-water. Reason is the enhanced thermal conductivity of nanofluid as compared to based fluid due to which it absorbs large amount of PV excess heat, bringing its temperature closer to ideal 25 ˚C.

Max. enhancement in PV-efficiency was about 32% in case of water-based PVT/PCM system as shown in Fig. 22 while when compared with Fig. 32, it can be observed that PV-efficiency can be increased by 60% using CNT-water nanofluid, when compared to conventional-PV. This shows effectiveness of nanofluids in enhancing PV performance due to their high thermal conductivity & heat absorbing characteristics.

Most of the researchers (Al-Waeli et al., 2017; Soltani et al., 2017; Hassani et al., 2016; Hasan et al., 2017; Al-Shamani et al., 2016)) found more than 50% enhancement in PV power output (see Fig. 33) by cooling with high heat absorbing nanofluids. Even about 160% enhancement was found by Al-Shamani et al. (2016) using SiC as nanofluid. Hence, nanofluids are highly recommended for active cooling of photovoltaics. Although nanofluids require power consumption to make them flow, but this can be easily compensated by efficiency enhancement and additional thermal power produced by heat absorption in nanofluid.

Overall efficiency is sum of electrical as well as thermal efficiency. With nanofluids, excess heat is highly extracted from PV, so temperature-dependent PV-efficiency increases significantly, while thermal energy of nanofluid can be further utilized, resulting in much enhanced overall efficiency of PVT system. Even more than 100% increase in overall efficiency is found by (Rejeb et al., 2016) using Cu-EG nanofluid (see Fig. 34).

3.3. Hybrid PVT system with nanofluid as optical filter

Jin and Jing (2017) prepared 2 stable, low cost and optimized magnetic ENFs i.e. Fe_3O_4/MB & Fe_3O_4/CS for Si & InGaP PV-cells, respectively by varying particle vol. fraction & electrolyte molar conc. When used for optical filtration in hybrid PV/T, results stated that MF values of ENFs for Si & InGaP PV-cells were higher than core/shell nanoparticle filter by 0.029 & 0.034 respectively. An et al. (2016) investigated performance of polypyrrole nanofluid by flowing it in quartz tube to absorb thermal energy in selective wavelengths and then passed through Al tubes over which PV-cells were pasted. PV-efficiency decreased from 20% to 12% with increase in nanoparticle conc. Overall efficiency enhanced by 13.3% at low conc. when compared to system without filter. Hassani et al. (2016) used Ag/water nanofluid as optical filter & absorber in hybrid PV/T with separate channels (D-1) & double pass (D-2) configurations for GaAs & Si PV-cells. D-1 performed better than D-2 due to higher heat absorption. As conc. increased, PV-

Table 5
Summary of proposed work by different researchers for PV cooling using nanofluid as coolant.

Ref no.	PV-panel specifications	Type of PVT system	Nature of Work and Duration	Nanofluid, Flowrate, Flow regime	Nanoparticle, Size & Concentration	Peak $T_{PV\text{-}ref}/T_{PVT}$	Key Findings
(Al-Waeli et al., 2017)	120 ± 3% W polycrystalline panel	PV-panel attached to nanofluid container	Outdoor experiment in hottest conditions, 1 day	SiC-water	SiC, 45–65 nm, 3 wt %	66 °C/-50 °C	PV temp. reduced by 16 °C. Electrical power & efficiency enhanced by 57% & 24.12% respectively.
(Sardarabadi and Passandideh-Fard, 2016)	40 W monocrystalline Si PV-panel	PV equipped with sheet & tube collector containing nanofluids	Outdoor experiment at Mashhad, Iran for some days in Aug & Sep	TiO_2-water, ZnO-water, Al_2O_3-water, 30 kg/h (Laminar flow)	TiO_2, 10–30 nm, 0.2 wt% - ZnO, 10–25 nm, 0.2 wt% - Al_2O_3, 20 nm, 0.2 wt%	TiO_2:-71 °C/-57 °C ZnO: 75 °C/-59 °C Al_2O_3: 71 °C/-58 °C	Max enhancement in PV-efficiency for TiO_2, ZnO, Al_2O_3/ water nanofluids was 6.54, 6.46 & 6.36% respectively.
(Hussien et al., n.d.)	Monocrystalline solar panels	PVT using nanofluid	Indoor experiment (24 mins) using sun simulator	Al_2O_3-water, 0.2L/s	Al_2O_3, 30 nm, 0.3 wt%	79.1 °C/42.2 °C	Cell temp. lowered by max 36.9 °C. PV-efficiency improved by 51.2%.
(Ghadiri et al., 2015)	Two 40 W monocrystalline Si PV-modules, 630 × 540 (mm)	PVT with nanofluid in flat plate collector	Indoor experiment with Solar simulator	Ferrofluid, 30L/h	Fe_3O_4, 45 nm, 1 wt % & 3 wt%	–	PV/T system performed better with 3 wt% conc. than 1 wt%. PV-efficiency enhanced by 37% using 3 wt% conc.
(Sardarabadi et al., 2014)	Two 40 W monocrystalline Si PV-modules	Nanofluid PVT system, sheet & tube collector	Outdoor experiment at Mashhad, Iran in July & August	SiO_2-water, 20, 30 & 40L/h	SiO_2, 11–14 nm, 1 wt% & 3 wt%	68 °C/-51 °C (1 wt %)	PV-efficiency increased from 10 to 11% by increasing conc. from 1 to 3 wt%.
(Hamdan and Kardasi, 2017)	Three 20 W monocrystalline panels (0.18 m^2)	PV cooling tested by 2 nanofluids	Outdoor experiment Jordan for 1 day	Al_2O_3-water - CuO-water	Al_2O_3, 0.1-.6 wt% CuO, 0.2-.7 wt%	(Al_2O_3) 47 °C/23 °C - (CuO) 49 °C/22 °C	Optimal conc. was 0.4%Al_2O_3 & 0.6%CuO for best PV performance as efficiency enhanced by 25 & 35%.
(Karami and Rahimi, 2014)	5 W polycrystalline PV-module, 310 × 190 (mm)	PV cooled by nanofluids channels on backside	Indoor experiment using solar simulator	Boehmite Nanofluid, (laminar flow up to 200 mL/min, turbulent from 200 to 400 mL/min)	$AlOOH{\cdot}xH_2O$, 5–10 nm, 0.01, 0.1 & 0.5 wt%	–	Helical channel showed better results than straight one. Performance order for different conc. was 0.1 > 0.1 > 0.5 wt%.
(Performance of cooling photovoltaic cells using nanofluids performance of cooling photovoltaic cells using nanofluids, 2016)	PV-module	PV cooled by channels on backside containing nanofluids	Numerical simulation	Boehmite Nanofluid, 80, 100, 200, 300 & 400 mL/min (Laminar flow up to 200, Turbulent after 200 mL/min)	$AlOOH{\cdot}xH_2O$, 5–10 nm, 0.01 wt %, 0.1 wt%, 0.5 wt %	–	Max. reduction in PV temp. of 24 °C & electrical power increase of 21.87% obtained using 0.1 wt% conc.
(Soltani et al., 2017)	5 W crystalline Si PV-cell, 30 × 15 (cm)	Hybrid PVT system using nanofluid	Outdoor experiment for 5 days	SiO_2-water - Fe_3O_4-water	SiO_2, 22 nm, 0.5 wt % - Fe_3O_4, 50 nm, 0.5 wt%	–	Electrical power increased by 54.29 & 52.40% with SiO_2 & Fe_3O_4 nanofluids respectively.
(Hassani et al., 2016)	Single Si crystalline	Nanofluids based PV/T system	Numerical simulation	CNT-water	CNT	147.7 °C/88.3 °C	Remarkable performance as PV temp. reduced by 59.4 °C resulting in 53.84% increase in electrical power & 61.5% increase in efficiency.
(Ebaid et al., 2018)	Three 50 W crystalline Si PV-cell	PV cooled by nanofluids in 23 parallel rectangular channels	Outdoor experiment at Jerash-Jordan for 1 day	Al_2O_3 in water -polyethylene glycol, TiO_2 in water-cetyltrimethylammonium bromide, 500–5000 mL/min (Laminar flow up to 1000 mL/min)	Al_2O_3, 0.01, 0.05 & 0.1 wt% - TiO_2, 0.01, 0.05, & 0.1 wt%	(0.01 wt% Al_2O_3 at 3000 mL/min) 57.5 °C/43 °C - (0.01 wt% TiO_2 at 3000 mL/min) 57.5 °C/44 °C	PV temp. lowered from 57.5 °C to 39.5 °C & 40.5 °C by Al_2O_3 & TiO_2 based nanofluids respectively at 0.01% conc. & 3000 mL/min flowrate.
(Hasan et al., 2017)	110 W polycrystalline Si PV-module, 1490 × 675 (mm)	PVT collector with jet impingement of nanofluid	Indoor experiment using Solar simulator	SiC, TiO_2 & SiO_2-water, 0.05, 0.067, 0.083, 0.10, 0.117, 0.133, 0.15, 0.167 (kg/s)	SiC, 1 wt% TiO_2, 1 wt% SiO_2, 1 wt%	(SiC) 87 °C/41 °C - (TiO_2) 87 °C/45 °C – (SiO_2) 87 °C/50 °C at 0.167 kg/s, 1000 W/m^2	Performance was max. for SiC followed by TiO_2 & SiO_2 at 0.167 kg/s. PV-efficiency enhanced by 60, 55 & 47% respectively.

(*continued on next page*)

Table 5 (*continued*)

Ref no.	PV-panel specifications	Type of PVT system	Nature of Work and Duration	Nanofluid, Flowrate, Flow regime	Nanoparticle, Size & Concentration	Peak $T_{PV\text{-}ref}/T_{PVT}$	Key Findings
(Sathieshkumar, 2017)	37 W monocrystalline PV-panel, 613 × 523 (mm)	PV connected with baffles, rectangular flow channel	Outdoor experiment at Tamilnadu, India for 1 day	$CuTiO_2$-water, 0.01, 0.02, 0.03 kg/s	$CuTiO_2$, 0.2 wt%	–	PV-efficiency increased by 13.83% at 0.02 kg/s.
(Michael and Iniyan, 2015)	37 W PV-module, 0.6 × 0.4 (m)	Nanofluid-based PV/T with Cu sheet instead of Tedlar layer	Outdoor experiment at Chennai in April	CuO-water, 0.01 kg/s (Laminar flow)	CuO, 0.05 vol%	with glazing: 63 °C/74.45 °C without glazing: 63 °C/57.74 °C	Overall efficiency enhanced by 33.6% (without glazing) when compared to ref-PV.
(Chandrasekar et al., 2013)	50 W monocrystalline PV, dimensions 600 × 600 (mm)	PV cooled by cotton wick structures, ends dipped in nanofluid	Outdoor experiment at Tiruchirappalli in April	CuO-water, Al_2O_3-water, 0 (no flow)	CuO, 0.1 vol% - Al_2O_3, 0.1 vol%	65 °C/59 °C (CuO/water) - 65 °C/54 °C (Al_2O_3/water)	Al_2O_3/water nanofluid showed better results than CuO/water as PV-efficiency enhanced by 5.55 & 7.78% respectively.
(Hussien and Noman, 2015)	Mono-crystalline Si PV-panel	PV with nanofluid in pipes at rear	Indoor experiment using Sun simulator	Al_2O_3-water, 0.2L/s	Al_2O_3, 30 nm, 0.1, 0.2, 0.3, 0.4, 0.5 vol %	(Water/nanofluid) 79.1 °C/42.2 °C	As compared to water-based PVT, max temp. drop of 36.9 °C & PV-efficiency increase of 51.25% found using 0.3 vol% conc.
(Al-Shamani et al., 2016)	120 W PV-panel	PVT with nanofluids in rectangular tube collector	Outdoor experiment in tropical climate of Malaysia	SiC-water, TiO_2-water, SiO_2-water, 0.068 to 0.170 kg/s	SiC, 1 wt% TiO_2, 1 wt% SiO_2, 1 wt%	(Water/SiC-water) 62.62 °C/47.51 °C - (Water/TiO_2-water) 62.62 °C/52.74 °C - (Water/SiO_2-water) 62.62 °C/58.95 °C at 0.068 kg/s	PVT performance enhanced by increasing mass flowrate from 0.068 to 0.170 kg/s. PV-efficiency enhancements were in order, SiC-water > TiO_2-water > SiO_2-water.
(Karami and Rahimi, 2014)	Monocrystalline Si PV-module, dimensions 18 × 50 (mm)	PV cooled by nanofluids in rectangular microchannel	Indoor experiment using Solar simulator	Boehmite Nanofluid, 0–300 mL/min (Laminar flow)	$AlOOH{\cdot}xH_2O$, 5–10 nm, 0.01 wt %, 0.1 wt%, 0.3 wt %	62.29 °C/32.5 °C	PVT performance enhanced by increasing flowrate from 0 to 300 mL/min. At higher conc., agglomeration occurred.
(Rejeb et al., 2016)	Monocrystalline Si PV-module, area 0.34 m^2	PVT with nanofluid in sheet & tube collector	Outdoor experiment & numerical simulation at Mashhad, Iran on 20/9/16	Cu-water Al_2O_3-water Cu-EG Al_2O_3-EG	Cu, 0.1 wt%, 0.2 wt %, 0.4 wt% - Al_2O_3, 0.1 wt%, 0.2 wt%, 0.4 wt%	–	Water-based nanofluids were more efficient than EG-based nanofluids. Cu nanoparticle enhanced efficiency more than Al_2O_3. Performance increased by increasing conc. from 0.1 to 0.4 wt%.

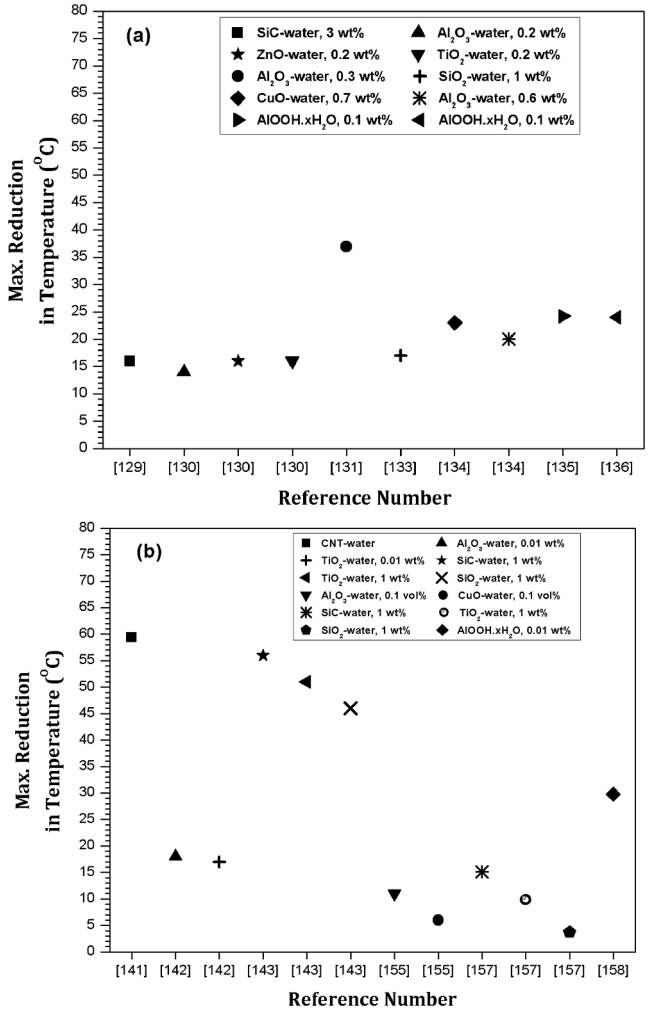


Fig. 31. Maximum reduction in PV-temperature using nanofluids as compared to conventional-PV achieved by different researchers in (a) and (b).

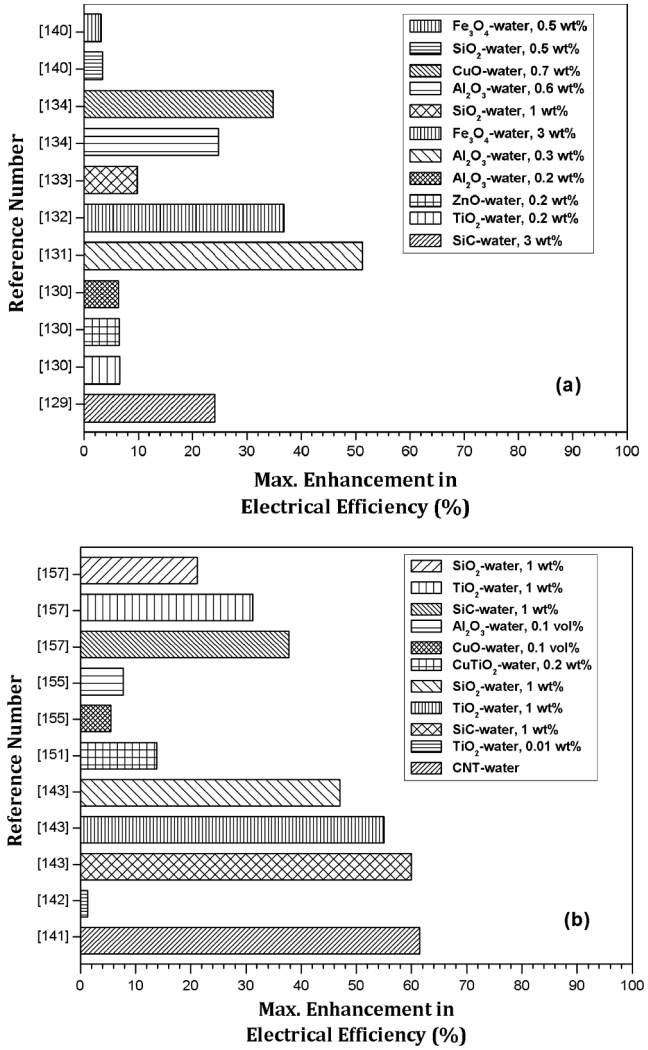


Fig. 32. Maximum enhancement in electrical efficiency using nanofluids as compared to conventional-PV achieved by different researchers in (a) and (b).

efficiency of D-1 decreased sharply than D-2. For GaAs PV-cell with D-1 configuration, max. PV & overall efficiencies were 10.9 & 30% respectively, at C = 160 & vol% = 1.5. For Si PV-cell, these were 6.2 & 24.4% respectively at C = 100 & vol% = 1.5. Jing et al. (2015) examined effect of particle size (5, 10, 25, 50 nm) & vol. fraction (0.5, 1, 2%) on optical filtration property of SiO_2-water nanofluids. (Fig. 35). Maximum transmittance (97%) was achieved with 5 nm particle size & 2% vol. fraction. Overall exergetic efficiency enhanced by 7% as compared to deionized water, by increasing concentration upto 40 & flow velocity upto 0.015 m/s.

Otanicar et al. (2013) compared nanoparticle-based & conventional thin film-based optical filters for use in hybrid PV/T system with 5 different PV-cell materials (Si, Ge, CdTe, InGaAs, InGaP). For all PV-cells except Ge, overall efficiency of later was although higher than the former due to better transmittance. Nanoparticle based filter required smaller thickness and could be easily tuned. Taylor and Otanicar (2015) examined effect of using nanofluid as optical filter & absorber in double pass configuration for hybrid PV/T system. After varying vol. fraction & channel thickness, 2 nanofluids (Ag & Cu/water) were prepared. PV-efficiency decreased due to filter imperfection, but overall efficiency enhanced by 18.6 & 8.1% for Ag & Cu/water nanofluids respectively, when compared to water-based system.

Cui and Zhu (2012) varied nanoparticles mass fraction & thickness of nanofluid (MgO/water) flow channel over PV-cell, and stated that sunlight transmittance decreased with increasing mass fraction & channel thickness. Maximum value was found with 0.02 wt% & 2 mm channel thickness. Overall efficiency increased from 15.9 to 61.9% while PV-efficiency decreased from 15.9 to 14.7% when compared to ref-PV. Saroha et al. (2015) experimentally determined effectiveness of 2 nanofluids (Ag & Au/water) as optical filter (10 mm channel thickness) & heat transfer media in hybrid PV/T system. PV-efficiency enhancements for Ag & Au/water nanofluids were 4.73 & 3.4% respectively, when compared to water-based system. Taylor et al. (2012) theoretically designed optimized nanofluid-based optical filters for hybrid PV/T systems with 5 different PV-cells (i.e. InGaP, CdTe, InGaAs, Si, Ge). Nanoparticles vol. fraction, filter thickness, type of base fluid, were varied for optimization. Although overall efficiency was lower than conventional pure fluid-optical filter but they were found more compact, inexpensive & controllable. Crisostomo et al. (2017) developed optical & heat loss model under different testing conditions i.e. different heat transfer fluids & concentration ratios for nanofluid-based optical filter for hybrid PV/T system. 4 nanofluids were prepared with conc. of Ag-SiO_2 nanoparticles varied as 0.025, 0.019, 0.013 & 0.006 wt%. Optical filter properties were not the best leading to lower PV-efficiency due to some imperfections but overall efficiency was higher than conventional-PV cells.

Taylor et al. (2013) designed and modelled nanofluid-based optical filters. When compared to conventional filters, these optical filters had slightly less transmittance & feasibility but their ability to easily get tuned according to system requirement made them good alternative. An

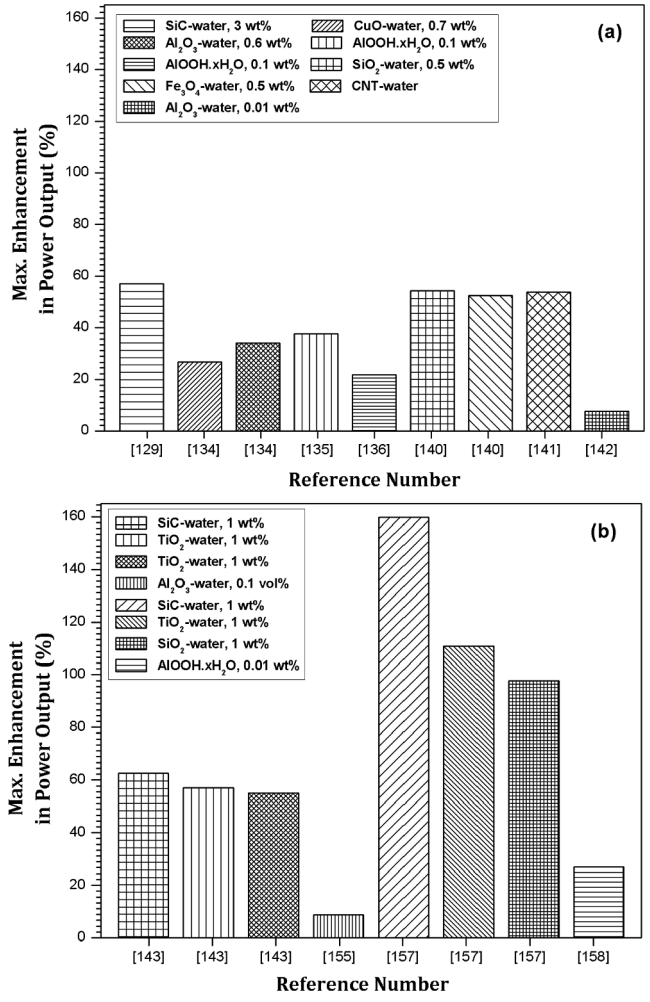


Fig. 33. Maximum enhancement in electrical power output using nanofluids as compared to conventional-PV achieved by different researchers in (a) and (b).

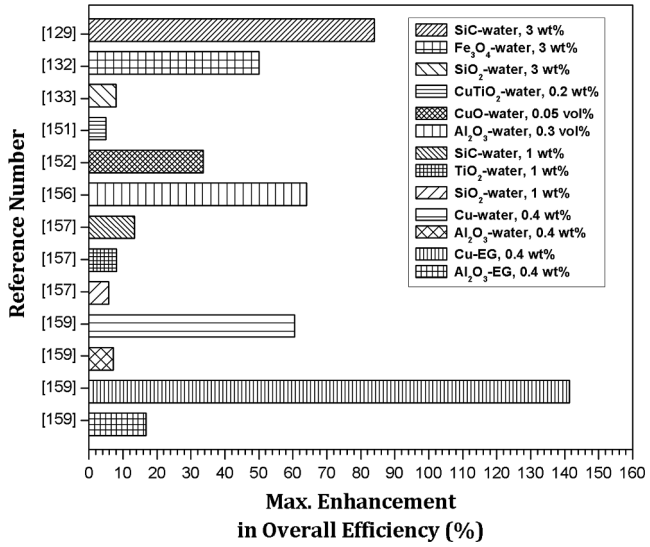


Fig. 34. Maximum enhancement in overall efficiency using nanofluids as compared to conventional-PV achieved by different researchers.

et al. (2016) used Cu_9S_5-oleylamine solution as optical filter for hybrid PVT system in which PV cooling was done by water beneath it. PV-efficiency increased with increasing conc. (0, 22.3, 44.6, 89.2 ppm) due to absorption of low-efficient solar radiation band. Maximum

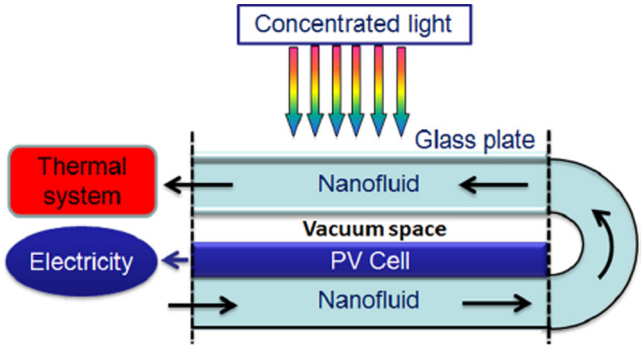


Fig. 35. Sketch of PV/T system concept with nanofluid flowing in single pass (Jing et al., 2015).

enhancement was 12.3% with 89.2 ppm. DeJarnette et al. (2015) designed a hybrid concentrating PVT system with nanofluid as optical filter. Au nanorods & ITO nanocrystals combinations were used in modelling to design tunable optical filter which could even work at 300 ˚C after which stability issues of nanofluid occur. The proposed system having cost lower than individual PV or CSP system, was found to have 27.8% overall efficiency.

High level of absorption with ability to tune solar spectrum was found in nanofluid-based optical filter by DeJarnette et al. (2014). Although transmittance was lower than conventional optical filters, but it needed smaller thickness in hybrid PVT system and these plasmonic nanoparticles could reduce cost significantly when used in double-pass architecture. Hjerrild et al. (2016) used nanofluid with multi-nanoparticles (Ag-SiO_2 nanodiscs & CNTs) for optical filtration of solar energy. Results stated that increasing concentration of Ag-SiO_2 increased merit function while CNTs provided absorption enhancement. Overall efficiency by using nanofluid, was higher than by base fluid.

Table 6 shows summary of proposed work by searchers for PV cooling using nanofluid as coolant.

3.4. Discussion

Although transmittance of solar light decreases as compared to water, but nanofluids are not bad transparent as transmittance obtained is more than 80% by all researchers (see Fig. 36). Maximum 95% transmittance is achieved by (Jing et al., 2015) using SiO_2-water nanofluid. Due to their relatively less thickness and ability to get tuned by varying wt%, nanofluids are good alternative for optical filter.

Fig. 37 shows about 10–20% enhancement in overall efficiency using nanofluid as optical filter as compared to that without filter. As nanofluids are tunable and require less thickness, so they can be used as optical filter for photovoltaics. Infra-red spectrum of sunlight is selectively absorbed by nanofluid before it falls on PV-surface. Although PV-efficiency is decreased but heat absorption in nanofluid before PV-surface as well as at rear, enhances overall efficiency of system.

4. Hybrid PVT system with combined PCM & nanofluid

A novel approach of comparing performances of 3 different systems i.e. conventional-PV, PVT with nanofluid, PVT/PCM with nanofluid through experimental investigation was proposed by Sardarabadi et al. (2017). Monocrystalline Si PV-panels of 40 W for all 3 systems were employed outdoor at Mashhad, Iran for some days in August & September. ZnO/water was used as nanofluid with 0.2 wt% conc. and it was flowed at 30 kg/h in Cu sheet & tube collector attached at PV-panel rear. Paraffin wax (MERCK) used as PCM having melting point in range of 42–72 ˚C, was placed in surroundings of tubes in third system. Maximum temperature reduction of about 7 ˚C & 16 ˚C were achieved for cases of PVT with nanofluid & PVT/PCM with nanofluid respectively, when compared with conventional-PV. PCM absorbed extra heat

Table 6
Summary of work proposed by different researchers for PV cooling using nanofluid as Spectral Filter.

Ref no.	PV-panel specifications	Type of PVT system	Nature of Work	Nanofluid, Layer thickness, Flowrate	Nanoparticle, Size & Concentration	Maximum Transmittance (%)	Key Findings
(Jin and Jing, 2017)	Si & InGaP PV-cells, dimensions, 1 × 1(m)	Hybrid PV/T with 2 types of ENFs as optical filters	–	Fe_3O_4/MB & Fe_3O_4/CS, 10 mm, 0.01 kg/s	Fe_3O_4, 20 nm	–	Vol. fraction & electrolyte molar conc. varied to prepare ENFs having required optical properties.
(An et al., 2016)	88 pieces of small cells (18 × 10 mm) connected in series	Hybrid PV/T with nanofluid filter pasted on Al tubes before poly-Si cells	Indoor experiment using Solar simulator	Polypyrrole nanofluid	C_4H_4NH, 13.68 nm, 0,3.12,6.25, 12.5 ppm	~78% with 3.12 ppm	PV-efficiency declined from 20 to 12% by increasing nanoparticle conc. With filter, overall efficiency increased by 13.3% at low conc.
(Hassani et al., 2016)	Si & GaAs PV-cell	Cascade nanofluid in PV/T system with optical & thermal properties	Numerical simulation	Ag-water, 20 mm (Si PV), 50 mm (GaAs PV), 0.08 kg/s	Ag, 10 nm, 0.003 vol% (Si PV) 0.0002 vol% (GaAs PV)	82%	Less thermal losses in separate channel configuration as compared to double pass. PV & overall efficiencies were 10.9 & 30% respectively at C = 160 & vol% = 1.5.
(Jing et al., 2015)	PV-module (100 monocrystalline cells), 1.56 × 1.56(m)	PV/T system with nanofluid as optical filter & absorber	Numerical simulation	SiO_2-water & 10 mm, (Turbulent flow)	SiO_2, 5,10,25,50 nm, 2 vol %	97%	Overall efficiency increased by max 7% as compared to deionized water at 40 conc. & 0.015 m/s flow velocity.
(Otanicar et al., 2013;5.)	5 different PV-cells i.e. Si, Ge, InGaP, CdTe, InGaAs	Hybrid PV/T system, nanofluid as optical filter & absorber	Numerical simulation	–	Al, Ag, Au, 60–104 nm	–	Overall efficiency decreased by using nanofluid as filter except for Ge. But it required small thickness & easily tuned.
(Taylor and Otanicar, 2015)	PV-module	Hybrid PV/T system, nanofluid as optical filter & absorber	Numerical simulation	Ag-water, Cu-water, 7 mm, 0.02 kg/s	Ag & Cu, 10 nm, 0.0001937 & 0.0000431 vol%	90%	Overall efficiency enhanced by 18.6 & 8.1% using Ag & Cu/water nanofluids respectively.
(Cui and Zhu, 2012)	25 W PV-panel	Hybrid PV/T system, nanofluid as optical filter & absorber	Indoor experiment using solar simulator	MgO-water, 8L/h	MgO, 10 nm, 0.02,0.06,0.1 wt%	~88% (0.02 wt%, 2 mm)	PV-efficiency declined from 15.9% to 14.7%, but overall efficiency of hybrid system increased from 15.9% to 61.9%.
(Saroha et al., 2015)	PV-module	Hybrid PV/T systems with nanofluid used as filter & absorber	Indoor experiment using Solar simulator	Ag-water, 10 mm - Au-water, 10 mm	Ag & Au, 5 nm, 0.00019 & 0.00025 vol%	–	Ag/water nanofluid showed higher enhancement in PV-efficiency than Au/water nanofluid (i.e. 4.7 & 3.4%).

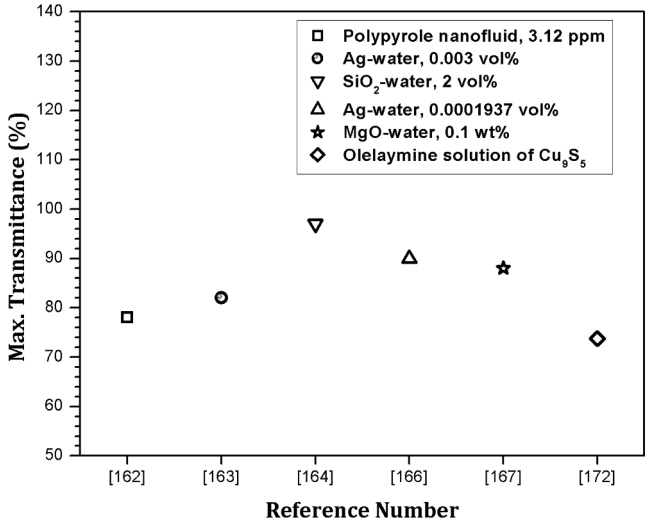


Fig. 36. Maximum transmittance of nanofluid as optical filter achieved by different researchers.

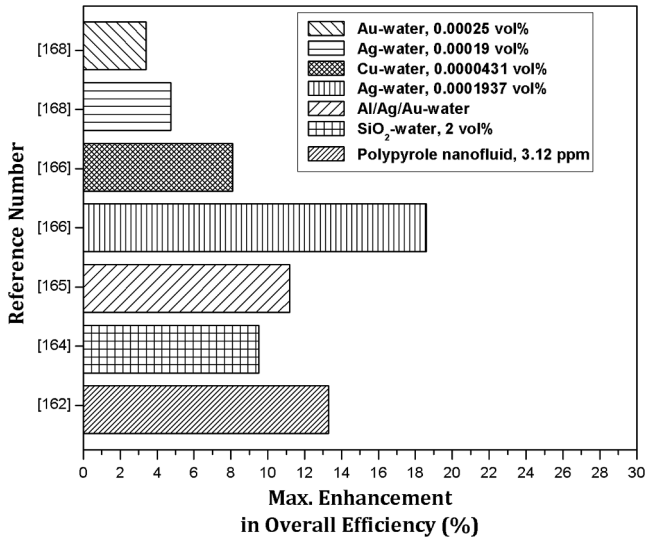


Fig. 37. Maximum enhancement in overall efficiency using nanofluid as optical filter as compared to that without filter achieved by different researchers.

of PV-panel in its latent heat. Hence it lowered and also produced stability in surface temperature of PV-panel. Moreover, ZnO/water nanofluid also absorbed high amount of heat due to its high value of thermal conductivity produced by nanoparticle addition. This heat absorption also resulted in increased thermal power. Electrical efficiency enhancements were 8 & 23% for PVT system with nanofluid & PVT/PCM system with nanofluid respectively when compared with conventional-PV. Hence, PVT/PCM with nanofluid performed better than other 2 systems.

Energy balance for control volume (PV-module and collector) was expressed as:

$$E_{in} = E_{el} + E_{th} + E_{st} + E_{loss}$$

where,

E_{in} = Incident irradiation coming from sun
E_{el} = Electrical power output = $V_{OC} * I_{SC} * FF$
E_{th} = Thermal power output = $m_f . C_{p,f} . (T_{f,o} - T_{f,i})$
E_{st} = Energy stored in PCM
E_{loss} = Energy loss

The overall efficiency of system was calculated as:

$$\eta_{pvt} = \frac{E_{th} + E_{el}}{E_{in}}$$

Al-Waeli et al. (2017) used nanofluid as well as nano-PCM for efficient cooling purpose of photovoltaics with focus on enhancing its electrical efficiency. SiC nanoparticles were added in working fluid (water) as well as PCM (paraffin wax). Nano-PCM was prepared in Ultrasonic shaker by investigating the addition of different mass fractions (0, 0.1, 0.5, 1, 2, 3, 4%) of SiC. At 3 wt%, thermal conductivity enhancement was maximum and so nano-PCM of 3 wt% conc. was used in the study. Similarly, nanofluid prepared also had 3% mass fraction of SiC in water. With increasing its flowrate in copper tubes, PV-surface temperature also reduced, so flowrate of 0.170 kg/s was used in the study. Nanoparticle addition increased heat transfer characteristics of both PCM and water. PCM first absorbed panel extra heat and then transferred it to nanofluid flowing in tubes leading to more heat extraction from module. By using a nanoparticle enhanced PCM and nanofluid based PVT, about 17 °C temperature reduction of PV cell was noted. Also, this system provided about 42.5% more heat gain when compared with simple PCM based PVT system used with water. Fig. 38. shows schematic diagram of experimental setup.

Table 7 shows summary of proposed work by different researchers for PV cooling using combination of PCM and nanofluid.

4.1. Hybrid PVT system with Hybrid-Nanofluid

Hybrid-nanofluids are the extension of mono-nanofluids and are obtained by hybrid-nanoparticles dispersion in base fluid. In most of cases, hybrid-nanofluids show higher thermal conductivity than nanofluids in most of cases. So, if employed in PVT system, they can enhance system's overall efficiency.

Sathieshkumar (2017) used hybrid nanofluid (0.2 wt% $CuTiO_2$-water) at varying flowrates (0.01, 0.02 and 0.03 kg/s) in baffle arrangement of collector for PV cooling. Maximum electrical and thermal efficiencies were found for hybrid nanofluid at 0.02 kg/s when compared with water-cooled & uncooled PV/T systems. Maximum electrical efficiency enhancements of 13.83% were determined for hybrid nanofluid as compared to conventional PV system. Hjerrild et al. (2016) used nanofluid with multi-nanoparticles (Ag-SiO_2 nanodiscs & CNTs) for optical filtration of solar energy to increase efficiency of hybrid PV/T system. Results stated that increasing concentration of Ag-SiO_2 increased the merit function while CNTs provided absorption enhancement, resulting in higher combined efficiency than base fluid. Younis et al. (2018) employed Al_2O_3-ZnO-H_2O nanofluid in hybrid PVT system. Due to low conc. (0.05 wt%) of nanoparticles, average PV efficiency increased very little i.e. less than 0.1% but overall efficiency increased by 4.1% on average due to its high thermal conductivity.

4.2. Discussion

The combined use of PCM & nanofluid is more effective approach for photovoltaics cooling than individual use of PCM or nanofluid. In this way, extra heat of PV-panel is extracted by 2 highly heat absorbing media i.e. first by PCM and then by nanofluid. Nanofluid, due to its higher thermal conductivity than base fluid, has higher heat transfer characteristics. Combining PCM with nanofluid not only lowers PV-surface temperature but also gives more uniformity in temperature due to its almost uniform contact with panel from rear. Moreover, addition of nanoparticles in PCM can enhance its thermal conductivity, thereby leading to its better heat absorption characteristics.

5. Environmental impacts

Keeping in front the present energy needs and potential solutions to meet these energy requirements, PV systems have found their great

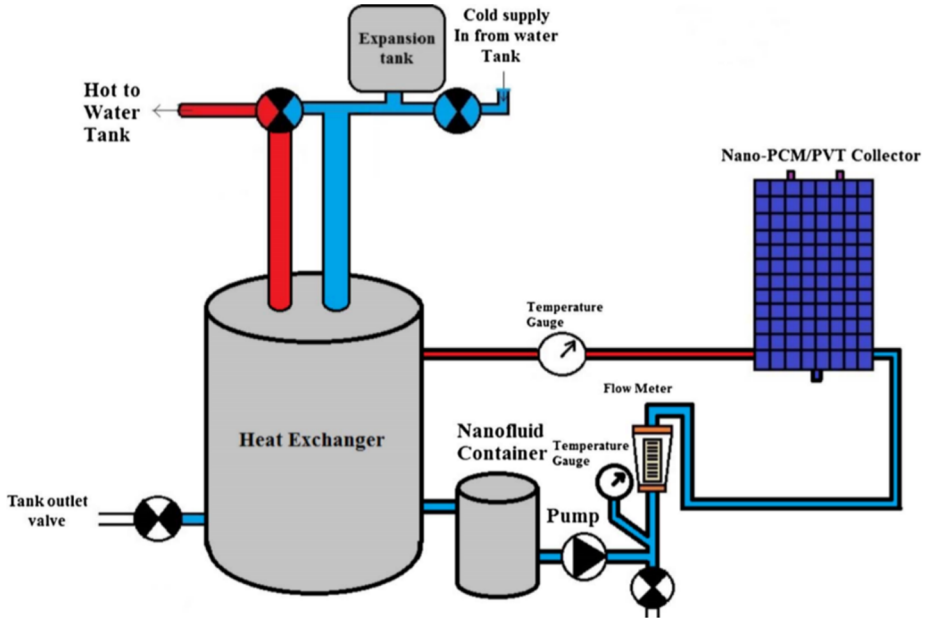


Fig. 38. Experimental setup – Schematic diagram (Al-Waeli et al., 2017).

importance. These systems when incorporated with PCMs, nanofluids or their combined use, can further decrease our dependency on fossil fuels for electricity generation. This can lead to reduced emissions of CO_2 and greenhouse gases.

Cellura et al. (2008) reported that PV/PCM systems are more efficient systems than conventional-PV systems, thereby producing more electrical energy. This could decrease factor of dependency on fossil fuels for electricity generation, leading to reduction in greenhouse gases and CO_2 emissions. Yang et al. (2018) also explained impact of increased electrical efficiency on reduction in fossil fuel consumption. Electrical efficiency of conventional-PV/T system enhanced from 6.98 to 8.16% with application of PCM. Increased power output indirectly meant reduced emissions of CO_2 and GHG due to decreased consumption of fossil fuels. Qiu et al. (2016) stated that PV/T systems based upon MPCM slurry have got a lot of potential in reducing fossil fuels consumption, thereby decreasing CO_2 emissions and so global warming in environment. Hassani et al. (2016) compared environmental impact of conventional-PV-module with some PV/T configurations having water and nanofluid as coolant & optical filter. CO_2 emissions were least for conventional-PV in first phase of life cycle but during operational phase, PV/T configuration with Ag/water nanofluid as coolant & optical filter, was most environment-friendly preventing emissions about 448 kg CO_2 eq m^{-2} yr^{-1}. According to Lari and Sahin (2017); nanofluid-based PV/T system could prevent 16974.57 tonnes CO_2 emissions into atmosphere.

Problem of global warming can be indirectly controlled to some extent by application of PCMs or nanofluids in PV systems. This would increase their electrical energy output which would decrease fossil fuels consumptions leading to reduction in CO_2 or GHG emissions.

6. Life-cycle cost and feasibility analysis – challenges

With growing interest in PCMs and nanofluids for thermal management leading to efficiency improvement of photovoltaics, cost & feasibility analyses of such systems find their great importance in this renewable energy era. Authors have done many studies in this regard although lack of data for complete analysis was found.

Hasan et al. (2010) determined cost-effectiveness of 5 different PCMs (i.e. RT20, C-L, C-P, $CaCl_2{\cdot}6H_2O$, SP22) used for photovoltaics cooling. In hot climates, only C-P and $CaCl_2{\cdot}6H_2O$ could make PV/PCM system financially viable while for other PCMs, reduction in PV-temperature wasn't enough to make them feasible. Nižetić et al. (2018) found insignificant difference in performance between pork fat & organic PCM used for PV cooling. Due to high cost and degradation of its physical properties in long-term, using pork fat as PCM was not found economically viable. Unit price of pork fat should be less than 0.5€/kg to make it favorable for consideration. Waqas et al. (2017;9.) reported that cost of conventional-PV was increased to $0.60 after modification with PCM. High cost was due to shipping cost as well as import tax duty. These could be excluded if cheap locally available PCM was used. Hasan et al. (2014) performed cost-analysis of PV/PCM system in hot (Vehari, Pakistan) & cool climate (Dublin, Ireland). PV/PCM was cost-effective in Vehari due to higher ambient temperatures while benefit of PV/PCM system was less than its incurred cost in cooler climate of Dublin. Klugmann-Radziemska and Wcisło-Kucharek (2017) performed economic-analysis of unmodified PV and PV/PCM systems for 30 years life cycle. During this period, PV/PCM system produced 2100 kW more power than unmodified PV system, thereby giving net profit of about 250€ after including PCM cost. This analysis was done using wholesale price of cheapest 42–44 paraffin employed to PV-module for cooling purpose. Japs et al. (2016) found that PV/PCM system showed promising results in terms of economic yield before noon only. But negative economic yield for EPEX spot market was observed for entire 30 days of measurement period. Hence, PV/PCM system didn't show economic viability.

Hasan et al. (2017) assessed PV/PCM system performance in hot climate of UAE. On yearly basis, about 5.5% increase in electrical power output was found. Cost-analysis was done by inclusion of PCM RT42 cost. Although economic benefit was 2.2 dollars/m^2, but 10 years payback period of proposed PV/PCM system was determined. It reduced to 3 years when low cost salt hydrate was used as PCM in analysis. A better economic scenario was expected if PCM cost decreased further due to increased interest in PV/PCM system in future. Yalçin and Öztürk (2013) examined that although integration of PCMs with PV could increase electrical energy output but due to additional cost of PCM & more than 20 years payback period, PV/PCM systems were not realistic. On contrary, PCM-based hybrid PVT systems could be feasible like BIPV in which PCM had double purpose i.e. higher electricity generation as well as indoor conditioning. Similar behavior of PCMs for PV thermal management was also presented by Smith et al. (2014) that single-junction PV/PCM systems are not cost-effective and that PCMs should be used in hybrid systems for their feasibility. Hjerrild et al.

Table 7
Summary of proposed work by different researchers for PV cooling using combination of PCM and nanofluid.

Ref no.	PV-panel specifications	Type of PV system	Nature of Work & Duration	Nanofluid Composition/ Flowrate	PCM tested, melting point (^{0}C)/Latent Heat (kJ/kg)	Peak $T_{PV\text{-}ref}$/ $T_{PVT\text{-}PCM}$	Key Findings
(Sardarabadi et al., 2017)	40 W monocrystalline PV-module, 630 × 540 (mm)	PVT with PCM & nanofluid	Outdoor experiment at Mashhad, Iran for August & September days	ZnO-water, 35–45 nm, 0.2 wt%/0.0083 kg/s	MERCK 107151, 42–72 °C/ 200–220 kJ/kg	61 °C/45 °C	Cell temp. dropped by 7 °C & 16 °C with nanofluid only and combined PCM & nanofluid respectively. PV-efficiency enhanced by 8% & 23% respectively.
(Al-Waeli et al., 2017)	PV-panel of 120 ± 3% W	PVT with nano-PCM & nanofluid	Outdoor experiment at Bangi, Malaysia for some sunny & clear days	SiC-water, 45–65 nm, 3 wt%/0.175 kg/s	Paraffin wax with SiC nanoparticles, 49 °C, 196 kJ/ kg	68.3 °C/39 °C	Cell temp. drop, electrical power & efficiency increase were 29.3 °C, 97.55% & 92.96% respectively.

(2016) stated that Ag-SiO_2 nanofluids are not very much expensive and they could reduce energy price if used in hybrid PVT system which could produce both electrical & thermal energy. Saroha et al. (2015) explained through experimental & numerical investigation results that nanofluid-based hybrid PV/T systems were feasible & viable. Hassani et al. (2016) suggested that optimized hybrid PV/T system in which nanofluid was used as coolant as well as optical filter, could supply electricity & thermal energy at a low cost. As optical filter, it selectively absorbed infra-red radiations to let PV-panel work at optimum solar heat conditions, while it cooled panel from rear and absorbed waste heat to store as thermal energy. According to Lari and Sahin (2017), silica/water nanofluid based hybrid PV/T system could bring 82% reduction in energy cost as compared to domestic cost of electricity in Saudi Arabia, with payback period of only 2 years.

Technical possibility & economic viability of PV/PCM system have been checked by many authors in past although that analysis is not very precise & accurate. Due to high material cost of PCM, shipping cost, tax import duty etc., these systems become expensive and are not feasible when operated in single junction due to long payback period. Their economic feasibility can be increased by integrating PCM in hybrid PV/T system, where extra PV heat absorbed by PCM is further utilized to gain thermal energy. In this way, system compactness can be increased with decrease in cost as well as payback period. Moreover, growing interest in such systems will be helpful in reducing cost of systems' components and materials leading to implementation of this efficient technology.

The difficulties in production process for nanofluids (Kouloulias et al., 2016), and higher cost due to requirement of advanced equipment hinder the applications of nanofluids (Keblinski et al., 2005; Pantzali et al., 2009; Gómez-Villarejo et al., 2017). Nanofluids have stability issues arising due to agglomeration (Ahmed Ali et al., 2016; Hashemi et al., 2014; Saidur et al., 2011; Khalil et al., 2017). Moreover, pumping power of nanofluids is increased due to viscosity which is higher than that of base fluids (Elsheikh et al., 2018). Hence, nanofluid-based PV/T systems are although efficient but are facing number of challenges which can be solved through technology advancements.

7. Conclusions and future recommendations

To counter efficiency decline problem of photovoltaics with rising temperature, a comprehensive review consisting of cooling techniques leading to efficiency enhancement, with PCMs, nanofluids and their combined effect has been presented in this study. Major conclusions of study are mentioned as follows:

- Solar energy incident on photovoltaics, not converted into electrical energy can be stored by PCMs in form of latent heat. This can help reduce PV-surface temperature by as much as 35 °C (see (Mahamudul et al., 2016; Stropnik and Stritih, 2016; Waqas and Ji, 2017). Moreover, by passive cooling with PCM, it is found that PV-efficiency can be enhanced up to 20% when compared to ref-PV. Effectiveness of PCM is more prominent in summer conditions than in winter conditions as PCM absorbs more heat in summer leading to more efficiency improvement than in winter.
- Selection of suitable PCM depends upon many parameters like ambient temperatures, solar intensity, latitude of location, wind velocity etc.
- Internally finned PV/PCM system is more effective than unfinned PV/PCM system as PCM contact with metallic fins helps in more rapid transfer of excess heat from PV-surface to PCM.
- Low thermal conductivity problem of PCMs can be solved quite effectively by addition of thermal conductivity enhancers like nanoparticles addition, mixing with graphite, inserting metal fiber structures etc. PV-surface temperature with compound PCM increases at a slower rate than pure-PCM after its melting, keeping cell temperature lower for longer durations.

- PV systems cooled with 2 PCMs at a time are more effective when compared to single PCM, for round the year performance due to better heat regulation & PV surface temperature uniformity.
- Hybrid PV/T systems are state of the art PV systems due to combined production of electrical & thermal power. Their performance can be further enhanced by integration with PCMs. This combination of passive & active cooling techniques helps in more lowering of PV-surface temperatures, leading to enhancement in PV-efficiency with extra thermal power generation.
- PV-efficiency of water-based hybrid PV/T system can be improved by as much as 32% when integrated with PCM (see (Zhou et al., 2017). Similarly, air-based hybrid PV/T systems with PCM can also provide ventilation of rooms along with increased PV-efficiency.
- Nanofluid-based hybrid PV/T systems are found to perform better than their base fluids due to enhanced thermal conductivity of nanofluid fluid, resulting in better heat transfer & absorbing characteristics. Maximum enhancement in PV-efficiency can reach more than 60% as compared to conventional-PV (see (Hassani et al., 2016).
- Overall efficiency can enhance by as much as 140% through nanofluid application. Maximum enhancement is found to be achieved with SiC-water and Cu-EG (see (Al-Waeli et al., 2017; Rejeb et al., 2016).
- Various parameters affecting performance of nanofluid-based hybrid PV/T systems include nanoparticle concentration, type of nanofluid, flow rate, inlet temperature etc. Effect of nanoparticle concentration is dependent upon type of fluid flow i.e. if flow is laminar, increasing conc. up to certain extent, reduces PV-surface temperature leading to efficiency enhancement while effect is reverse, if flow is turbulent. Effect of nanoparticle diameter on nanofluids performance for PV cooling is not much significant as from available literature.
- Although nanofluids transmittance is not as good as conventional fluids but due to their relatively smaller thickness and ability to tune by varying weight concentration, they can be considered good alternative for use as spectral filter.
- Double pass configuration of nanofluids when used as spectral filter & coolant, is more effective than single pass configuration due to more heat extraction and less thermal losses.
- Combined use of PCM & nanofluid is more effective approach for PV cooling than by individual use of PCM or nanofluid as extra heat of PV-panel is extracted by 2 highly heat absorbing media i.e. first by PCM and then by nanofluid. Combining PCM with nanofluid not only lowers PV-surface temperature but also gives more temperature uniformity due to uniform contact of PCM with panel. If combination is made between nanofluid & nano-PCM, better thermal energy management can be achived (see (Al-Waeli et al., 2017).
- PV systems when cooled with above mentioned methods, can decrease dependency on fossil fuels for electricity generation and reduces CO_2 & greenhouse gases emissions. Nanofluid-based PV/T system could prevent 16,974.57 tones CO_2 emissions into atmosphere (see (Lari and Sahin, 2017).
- Due to high PCM cost, PV/PCM systems mostly become expensive & infeasible when operated in single junction as payback period can reach 20 years (see (Yalçin and Öztürk, 2013). Among various PCMs, only salt hydrates like C-P, $CaCl_2{\cdot}6H_2O$ etc. can make PV/PCM system financially viable in hot climates (see (Hasan et al., 2010). Economic feasibility can be increased by integrating PCM with hybrid PV/T system where extra PV heat absorbed by PCM is further utilized to gain thermal energy. In this way, system compactness is also increased with decrease in cost as well as payback period.
- Nanofluid-based PV/T systems are although very efficient systems but they are facing quite a number of challenges like high cost of active equipment, stability issues due to agglomeration, which needs a serious consideration in near future.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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